

Exercise J02

Q1. Find the equation of the ellipse with its centre (1, 2), focus at (6, 2) and passing through the point (4, 6).

Ans. $20x^2 + 45y^2 - 40x - 180y - 700 = 0$

Sol. $\frac{(x-1)^2}{a^2} + \frac{(y-2)^2}{b^2} = 1$

It passes through (4, 6)

$$\frac{9}{a^2} + \frac{16}{b^2} = 1 \text{ and } ae = 5$$

$$\Rightarrow b^2 = a^2(1 - e^2) \Rightarrow b^2 = a^2 - a^2e^2 \Rightarrow b^2 = a^2 - 25$$

$$\Rightarrow \frac{9}{a^2} + \frac{16}{a^2 - 25} = 1 \Rightarrow (a^2 - 5)(a^2 - 45) = 0 \Rightarrow \text{either } a^2 = 5 \text{ or } a^2 = 45$$

$$\text{but } a^2 \neq 5 \quad \therefore a^2 = 45 \Rightarrow b^2 = 20$$

Q2. An ellipse passes through the points (-3, 1) & (2, -2) & its principal axis are along the coordinate axes in order. Find its equation.

Ans. $3x^2 + 5y^2 = 32$

Sol. Let equation $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$

passing (-3, 1) $\frac{9}{a^2} + \frac{1}{b^2} = 1$ (1)

(2, -2) $\frac{4}{a^2} + \frac{4}{b^2} = 1$ (2)

from (1) & (2)

$$a^2 = \frac{32}{3}, b^2 = \frac{32}{5}$$

$$3x^2 + 5y^2 = 32$$

Q3. 'O' is the origin & also the centre of two concentric circles having radii of the inner & the outer circle as 'a' & 'b' respectively. A line OPQ is drawn to cut the inner circle in P & the outer circle in Q. PR is drawn parallel to the y-axis & QR is drawn parallel to the x-axis. Prove that the locus of R is an ellipse touching the two circles. If the foci of this ellipse lie on the inner circle, find the ratio of inner : outer radii & find also the eccentricity of the ellipse.

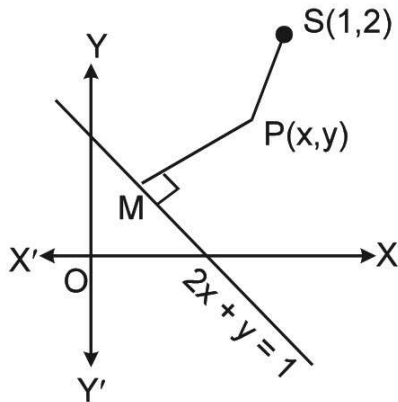
Ans. $\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}}$

Sol. -

Q4. Find the equation to the hyperbola whose directrix is $2x + y = 1$, focus $(1, 1)$ & eccentricity $\sqrt{3}$. Find also the length of its latus rectum.

Ans. -

Sol.



Let $P(x, y)$ be any point on the hyperbola.

Draw PM perpendicular from P on the directrix.

Then by definition $SP = e PM$

$$\Rightarrow (SP)^2 = e^2 (PM)^2$$

$$\Rightarrow (x - 1)^2 + (y - 2)^2 = 3 \left\{ \frac{2x + y - 1}{\sqrt{4 + 1}} \right\}^2$$

$$\Rightarrow 5(x^2 + y^2 - 2x - 4y + 5) = 3(4x^2 + y^2 + 1 + 4xy - 2y - 4x)$$

$$\Rightarrow 7x^2 - 2y^2 + 12xy - 2x + 14y - 22 = 0$$

which is the required hyperbola.

Q5. The hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ passes through the point of intersection of the lines, $7x + 13y - 87 = 0$ and $5x - 8y + 7 = 0$ & the latus rectum is $32\sqrt{2}/5$. Find 'a' & 'b'.

Ans. $a^2 = 25/2$; $b^2 = 16$

Sol. Point of intersection of lines

$7x + 13y - 87 = 0$ & $5x - 8y + 7 = 0$ is $(5, 4)$.

$$\text{Then } \frac{25}{a^2} - \frac{16}{b^2} = 1 \quad \dots(i)$$

$$\text{Also latus rectum LR} = \frac{2b^2}{a} = \frac{32\sqrt{2}}{5}$$

$$\Rightarrow b^2 = \frac{16\sqrt{2}a}{5} \quad \dots(ii)$$

$$\text{From (i) \& (ii) } a^2 = \frac{25}{2}, b^2 = 16$$

- Q6. For the hyperbola $\frac{x^2}{100} - \frac{y^2}{25} = 1$ prove that
 (i) eccentricity = $\sqrt{5}/2$
 (ii) SA. S'A = 25, where S & S' are the foci & A is the vertex.

Ans. -

Sol. $\frac{x^2}{10^2} - \frac{y^2}{5^2} = 1 \Rightarrow e = \sqrt{1 + \frac{25}{100}} = \frac{\sqrt{5}}{2}$

SA = ae - a

S'A = ae + a $\Rightarrow$ SA.S'A = $a^2(e^2 - 1) = 25$.

- Q7. Find the centre, the foci, the directrices, the length of the latus rectum, the length & the equations of the axes of the hyperbola $16x^2 - 9y^2 + 32x + 36y - 164 = 0$.

Ans. $(-1, 2)$; $(4, 2)$ & $(-6, 2)$; $5x - 4 = 0$ & $5x + 14 = 0$; $\frac{32}{3}$; 6 ; 8 ; $y - 2 = 0$; $x + 1 = 0$

Sol. Given hyperbola can be written as

$$\frac{(x+1)^2}{9} - \frac{(y-2)^2}{16} = 1$$

so $e = \frac{5}{3}$, centre is $(-1, 2)$

foci = $(-1 \pm 5, 2) = (-6, 2)$ & $(4, 2)$

directrix is $x + 1 = \frac{9}{5} \Rightarrow x = -1 \frac{9}{5}$

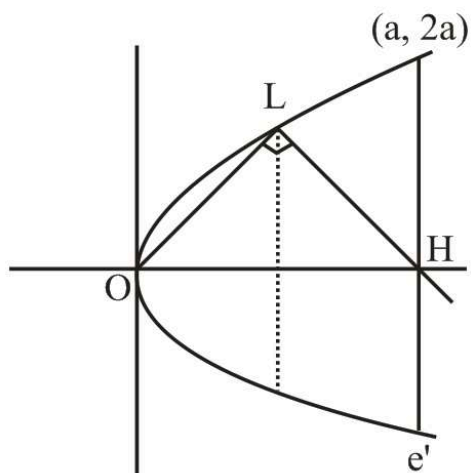
L.R. = $\frac{32}{3}$, Length of axes is 8 and 6

Equation of axis is $y - 2 = 0$ and $x + 1 = 0$.

- Q8. 'O' is the vertex of the parabola $y^2 = 4ax$ & L is the upper end of the latus rectum. If LH is drawn perpendicular to OL meeting OX in H, prove that the length of the double ordinate through H is $4a\sqrt{5}$.

Ans. -

Sol.



$$y^2 = 4ax$$

$$MOL = 2$$

$$MLH = -\frac{1}{2}$$

$$y - 2a = -\frac{1}{2}(x - a)$$

$$H(5a, 0)$$

$$y^2 = 4a(5a)$$

$$y = \pm 2a\sqrt{5}$$

$$ec' = 4a\sqrt{5}$$

Q9. Suppose x and y are real numbers and that $x^2 + 9y^2 - 4x + 6y + 4 = 0$ then find the maximum value of $(4x - 9y)$.

Ans. -

Sol. $x^2 - 4x + 4(3y)^2 + 6y + 1 = 1$

$$(x - 2)^2 + (3y + 1)^2 = 1$$

$$(x - 2)^2 + \frac{\left(y + \frac{1}{3}\right)^2}{\frac{1}{9}} = 1$$

$$x - 2 = \cos \theta \qquad y + \frac{1}{3} = \frac{1}{3} \sin \theta$$

$$x = 2 + \cos \theta \qquad y = \frac{1}{3} \sin \theta - \frac{1}{3}$$

$$z = 4x - 9y$$

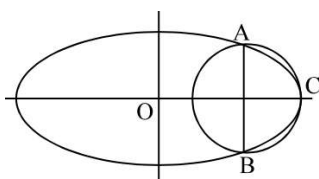
$$= 8 + 4 \cos \theta + 3 \sin \theta - 3$$

$$z = 5 + 4 \cos \theta + 3 \sin \theta$$

$$= 5 + [-6, 5]$$

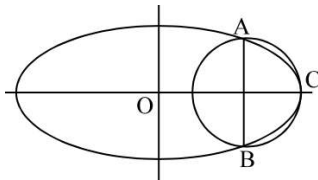
$$z_{max} = 5 + 5 = 10$$

Q10. A circle intersects an ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ precisely at three points A, B, C as shown in the figure. AB is a diameter of the circle and is perpendicular to the major axis of the ellipse. If the eccentricity of the ellipse is $4/5$, find the length of the diameter AB in terms of a .



Ans. $\frac{18a}{17}$

Sol.



Let radius of circle is ' r '

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$$

$$e = \sqrt{1 - \frac{b^2}{a^2}}$$

$$\frac{16}{25} = 1 - \frac{b^2}{a^2}$$

$$\frac{b^2}{a^2} = \frac{9}{25}$$

let $A(a \cos \theta, b \sin \theta)$

$$AB = 2b \sin \theta = 2r \quad \dots\dots(1)$$

$$r = b \sin \theta$$

$$a = a \cos \theta + r \quad \dots\dots(2)$$

from (1) & (2)

$$a(1 - \cos \theta) = b \sin \theta$$

$$\tan \frac{\theta}{2} = \frac{b}{a} = \frac{3}{5}$$

$$\sin \theta = \frac{2 \cdot \frac{3}{5}}{1 + \frac{9}{25}} = \frac{15}{17}$$

$$2r = \left(\frac{3}{5} a \cdot \left(\frac{15}{17} \right) \right)^2$$

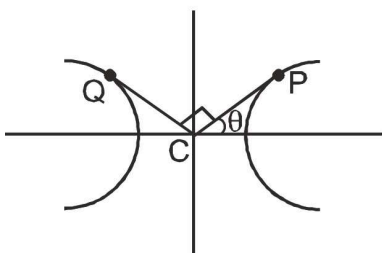
$$= \left(\frac{99}{17} \right)^2$$

$$= 189$$

Q11. If two points P & Q on the hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ whose centre is C be such that CP is perpendicular to CQ & $a < b$, then prove that $\frac{1}{CP^2} + \frac{1}{CQ^2} = \frac{1}{a^2} - \frac{1}{b^2}$.

Ans. -

Sol.



$$CP = \frac{x-0}{\cos \theta} = \frac{y-0}{\sin \theta} = r_1 \text{ where } CP = r_1$$

$$\therefore P(r_1 \cos \theta, r_1 \sin \theta)$$

$$\text{Similarly } Q\left(r_2 \cos\left(\frac{\pi}{2} + \theta\right), r_2 \sin\left(\frac{\pi}{2} + \theta\right)\right)$$

$$Q(-r_2 \sin \theta, r_2 \cos \theta)$$

P & Q lies on Hyperbola

$$\therefore r_1^2 \left(\frac{\cos^2 \theta}{a^2} - \frac{\sin^2 \theta}{b^2} \right) = 1$$

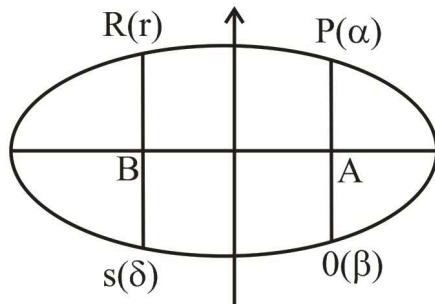
$$\therefore r_1^2 = \frac{a^2 b^2}{(b^2 \cos^2 \theta - a^2 \sin^2 \theta)} \quad \& \quad r_2^2 = \frac{a^2 b^2}{(b^2 \sin^2 \theta - a^2 \cos^2 \theta)}$$

$$\therefore \frac{1}{r_1^2} + \frac{1}{r_2^2} = \frac{b^2 - a^2}{a^2 b^2} = \frac{a}{a^2} - \frac{1}{b^2} \text{ H.P.}$$

Q12. If any two chords be drawn through two points on the major axis of an ellipse equidistant from the centre, show that $\tan \frac{\alpha}{2} \cdot \tan \frac{\beta}{2} \tan \frac{\gamma}{2} \cdot \tan \frac{\delta}{2} = 1$, where $\alpha, \beta, \gamma, \delta$ are the eccentric angles of the extremities of the chords.

Ans. -

Sol.



$$\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$$

$$A(\alpha, 0) \quad B(-\alpha, 0)$$

equation of chord PQ

$$\frac{x}{a} \cos\left(\frac{\alpha + \beta}{2}\right) + \frac{y}{b} \sin\left(\frac{\alpha + \beta}{2}\right) = \cos\left(\frac{\alpha - \beta}{2}\right)$$

Passing $(\alpha, 0)$

$$\frac{d}{a} \cos\left(\frac{\alpha + \beta}{2}\right) = \cos\left(\frac{\alpha - \beta}{2}\right)$$

$$\frac{d}{a} = \frac{\cos\left(\frac{\alpha - \beta}{2}\right)}{\cos\left(\frac{\alpha + \beta}{2}\right)}$$

$$\frac{d - a}{d + a} = \frac{\cos\left(\frac{\alpha - \beta}{2}\right) - \cos\left(\frac{\alpha + \beta}{2}\right)}{\cos\left(\frac{\alpha - \beta}{2}\right) + \cos\left(\frac{\alpha + \beta}{2}\right)}$$

$$\left(\frac{d-a}{d+a}\right) = \tan\left(\frac{\alpha}{2}\right) \cdot \tan\left(\frac{\beta}{2}\right) \quad \dots\dots(1)$$

similaly RS passing through $(-d, 0)$

$$\left(\frac{-d-a}{-d+a}\right) = \tan\left(\frac{r}{2}\right) \cdot \tan\left(\frac{\delta}{2}\right) \quad \dots\dots(2)$$

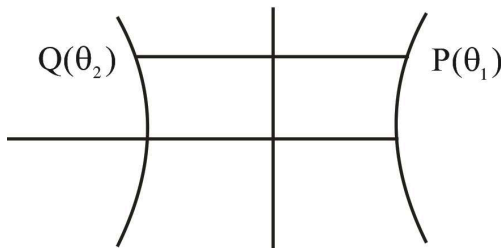
$$(1) \times (2)$$

$$\tan\left(\frac{\alpha}{2}\right) \cdot \tan\left(\frac{\beta}{2}\right) \cdot \tan\left(\frac{r}{2}\right) \cdot \tan\left(\frac{\delta}{2}\right) = 1$$

Q13. If θ_1 & θ_2 are the parameters of the extremities of a chord through $(ae, 0)$ of a hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$, then show that $\tan \frac{\theta_1}{2} \cdot \tan \frac{\theta_2}{2} + \frac{e-1}{e+1} = 0$.

Ans. -

Sol.



$$\begin{aligned} & \frac{x}{a} \cos\left(\frac{\theta_1 - \theta_2}{2}\right) - \frac{y}{b} \sin\left(\frac{\theta_1 + \theta_2}{2}\right) \\ &= \cos\left(\frac{\theta_1 + \theta_2}{2}\right) \end{aligned}$$

passing $(ae, 0)$

$$e = \frac{\cos\left(\frac{\theta_1 + \theta_2}{2}\right)}{\cos\left(\frac{\theta_1 - \theta_2}{2}\right)}$$

$$\frac{e-1}{e+1} = \frac{\cos\left(\frac{\theta_1 + \theta_2}{2}\right) - \cos\left(\frac{\theta_1 - \theta_2}{2}\right)}{\cos\left(\frac{\theta_1 + \theta_2}{2}\right) + \cos\left(\frac{\theta_1 - \theta_2}{2}\right)}$$

$$\frac{e-1}{e+1} = -\tan\left(\frac{\theta_1}{2}\right) \cdot \tan\left(\frac{\theta_2}{2}\right)$$

Q14. Find the equations of the chords of the parabola $y^2 = 4ax$ which pass through the point $(-6a, 0)$ and which subtends an angle of 45° at the vertex.

Ans. $y \pm 2(x + 6a) = 0$

Sol. Let slope of chord is m

$$y - 0 = m(x + 6a)$$

$$mx - y + 6a = 0$$

$$\frac{mn - y}{-6am} = 1 \quad = \text{HF}$$

$$y^2 - 4an = 0$$

$$y^2 - 4an\left(\frac{mn - y}{-6am}\right) = 0$$

$$y^2 + \frac{2x}{3m}(mn - y) = 0$$

$$3my^2 + 2mn^2 - 2xy = 0$$

$$\tan \theta = \left[\frac{2\sqrt{h^2 - ab}}{a + b} \right]$$

$$\tan 45 = \frac{2\sqrt{1 - 6m^2}}{5m} \Rightarrow 25m^2 = 4(1 - 6m^2)$$

$$49m^2 = 4$$

$$n = \pm \frac{2}{7}$$

$$y = \pm \frac{2}{7}(n + 6a)$$

Q15. Tangents drawn from the point $P(2, 3)$ to the circle $x^2 + y^2 - 8x + 6y + 1 = 0$ touch the circle at the points A and B . The circumcircle of the $\triangle PAB$ cuts the director circle of ellipse $\frac{(x+5)^2}{9} + \frac{(y-3)^2}{b^2} = 1$ orthogonally. Find the value of b^2 .

Ans. 54

Sol. $P(2, 3)$ $x^2 + y^2 - 8x + 6y + 1 = 0$

$$e(4, -3)$$

circum circle of $\triangle PAB$

$$(x - 2)(x - 4) + (y - 3)(y + 3) = 0$$

$$x^2 - 6x + 8 + y^2 - a = 0$$

$$x^2 = y^2 - 6x - 1 = 0 \quad \dots(1)$$

Director circle of ellipse

$$(x + 5)^2 + (y - 3)^2 = a + b^2$$

$$x^2 + 10x + y^2 - 6y + 25 = b^2$$

$$x^2 + y^2 + 10x - 6y + 25 - b^2 = 0 \quad \dots(2)$$

(1) & (2) are arthogonal to each other

$$2g_1g_2 + 2f_1f_2 = c_1 + c_2$$

$$2(3)(-5) + 2(0)(3) = -1 + 25 - b^2$$

$$b^2 = 54$$

Q16. Tangents are drawn from the point (α, β) to the hyperbola $3x^2 - 2y^2 = 6$ and are inclined at angles θ and ϕ to the x-axis. If $\tan \theta \cdot \tan \phi = 2$, prove that $\beta^2 = 2\alpha^2 - 7$.

Ans. -

Sol. $3x^2 - 2y^2 = 6$

$$\frac{x^2}{2} - \frac{y^2}{3} = 1$$

Let the equation of tangent

$$y = mx + \sqrt{a^2m^2 - b^2}$$

passes through (α, β)

$$(\beta - m\alpha)^2 = a^2m^2 - b^2$$

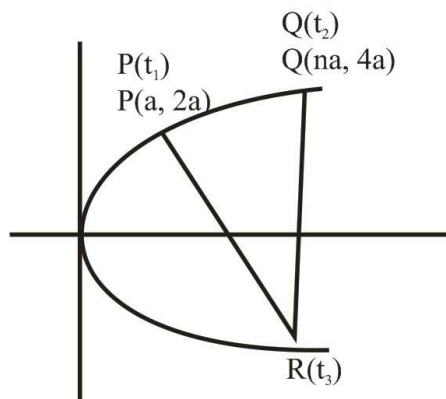
$$m^2\alpha^2 + \beta^2 - 2m\alpha\beta = a^2m^2 - b^2$$

$$m^2(\alpha^2 - a^2) - 2m\alpha\beta + \beta^2 + b^2 = 0$$

Q17. Show that the normals at the points $(4a, 4a)$ & at the upper end of the latus rectum of the parabola $y^2 = 4ax$ intersect on the same parabola.

Ans. -

Sol.



$$t_1 = 1, t_2 = 2$$

for PR

$$t_3 - t_1 = \frac{2}{t_1}$$

$$= -1 - 2 = -3$$

for QR

$$t_3 = -t_2 - \frac{2}{t_2}$$

$$= -2 - \frac{2}{2} = -3$$

Q18. Prove that the locus of the middle point of portion of a normal to $y^2 = 4ax$ intercepted between the curve & the axis is another parabola. Find the vertex & the latus rectum

of the second parabola.

Ans. $(a, 0)$; a

Sol. Equation of normal at $(am^2, 2am)$

on $y^2 = 4ax$

$$y + mx = 2am + am^3 \quad \dots (1)$$

It cuts x - axis at $y = 0$ i.e. $(2a + am^2, 0)$

Let middle point (h, k)

$$2h = am^2 + 2a + am^2$$

$$h = am^2 + a \text{ \& } k = am \quad \dots (2)$$

from (1) & (2)

$$h = a \frac{k^2}{a^2} + a$$

$$\text{Locus } y^2 = a(x - a)$$

$$\text{vertex } (a, 0) \quad \text{L.R.} = a$$

Q19. Find the equations of the tangents to the parabola $y^2 = 16x$, which are parallel & perpendicular respectively to the line $2x - y + 5 = 0$. Find also the coordinates of their points of contact.

Ans. $2x - y + 2 = 0, (1, 4)$; $x + 2y + 16 = 0, (16, -16)$

Sol. $y^2 = 16x$

tangent to parabola perpendicular to $2x - y + 5 = 0$

$$M_L = 2$$

$$M_T = \frac{-1}{2}$$

$$y = mn + \frac{9}{m}$$

$$y = \frac{-x}{2} + \frac{4}{(-\frac{1}{2})}$$

$$y = \frac{-x}{2} - 8$$

$$2y = -x + 16$$

$$x + 2y = -16 \quad \dots\dots\dots (1)$$

let P.O.C (x_1, y_1)

$$yy_1 = 8(x + x_1)$$

$$8x - yy_1 + 8x_1 = 0 \quad \dots\dots\dots (2)$$

$$\frac{8}{2} = \frac{-y_1}{-1} = \frac{8x_1}{2}$$

$$y_1 = -16, \quad x_1 = 16$$

$$(x_1, y_1) = (16, -16)$$

tangent parallel to $2x - y + 5 = 0$

$$m_T = m_1 = 2$$

$$y = 2x + \frac{4}{2}$$

$$2x - y + 2 = 0 \quad \dots\dots\dots (3)$$

compair with (2)

$$\frac{8}{2} = \frac{-y_1}{-1} = \frac{8x_1}{2}$$

$$y_1 = 4, \quad x_1 = 1$$

Q20. Find the equations of the tangents of the parabola $y^2 = 12x$, which passes through the point (2,5). Also find the point of contact.

Ans. $3x - 2y + 4 = 0$; $x - y + 3 = 0$

Sol. $y^2 = 12x$

tangent

$$y = mx + \frac{a}{n}$$

passing (2, 5)

$$5 = 2n + \frac{3}{n}$$

$$2n^2 - 5n + 3 = 0$$

$$2m^2 - 2n - 3m + 3 = 0$$

$$2n(n - 1) - 3(m - 1) = 0$$

$$(2m - 3)(n - 1) = 0$$

$$m = 1, \quad m = \frac{3}{2}$$

$$y = x + 3$$

$$y = \frac{3}{2}x + \frac{3}{\frac{3}{2}}$$

$$x - y + 3 = 0 \quad \dots\dots (i)$$

$$2y = 3x + 4$$

$$3x - 2y + 4 = 0 \quad \dots\dots (2)$$

let P.O.C $P(x_1, y_1)$

$$yy_1 = 6(x + x_1)$$

$$6x - yy_1 + 6x_1 = 0 \quad \dots (3)$$

compair with (1)

$$\frac{6}{1} = \frac{-y_1}{-1} = \frac{6x_1}{3}$$

$$(3, 5)$$

compair with (2)

$$\frac{6}{3} = \frac{y_1}{-2} = \frac{6x_1}{4}$$

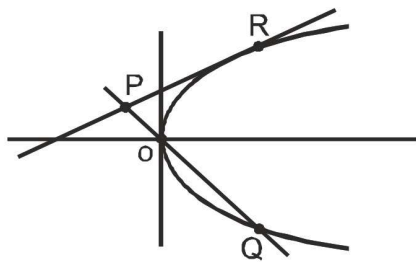
$$y_1 = 4, \quad x_1 = \frac{4}{3}$$

$$\left(\frac{4}{3}, 4\right)$$

Q21. Through the vertex O of the parabola $y^2 = 4ax$, a perpendicular is drawn to any tangent meeting it at P & the parabola at Q. Show that $OP \cdot OQ = \text{constant}$.

Ans. 16

Sol.



$$y = mx + \frac{a}{m} \quad \dots (i)$$

equation of OP is

$$y = -\frac{1}{m}x$$

$$OP = \frac{\frac{a}{m}}{\sqrt{1+m^2}}$$

equation (ii) meets the parabola at Q

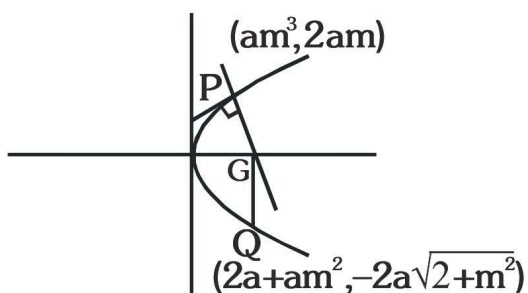
$$\frac{1}{m^2}x^2 = 4ax \Rightarrow x = 4am^2, y = -4am$$

$$\therefore OQ = 4am\sqrt{1+m^2}, \quad OP \cdot OQ = 4a^2$$

Q22. The normal at a point P to the parabola $y^2 = 4ax$ meets its axis at G. Q is another point on the parabola such that QG is perpendicular to the axis of the parabola. Prove that $QG^2 - PG^2 = \text{constant}$.

Ans. -

Sol.



Normal at $P(am^2, 2am)$ on $y^2 = 4ax$

$$y + mx = 2am + am^3 \quad \dots (1)$$

$G(2a + am^2, 0)$

Equation of QG is $x = 2a + am^2$

Solving with parabola we get

$$y = 2a\sqrt{2 + m^2}$$

$$QG^2 - PG^2$$

$$= 4a^2(2 + m^2) - (am^2 - am^2 - 2a)^2 - (2am)^2$$

$$= 4a^2 \text{ which is constant}$$

Q23. If the normal at $P(18, 12)$ to the parabola $y^2 = 8x$ cuts it again at Q , show that $9PQ = 80\sqrt{10}$

Ans. -

Sol. $y^2 = 8x$

$$P(18, 12)$$

$$P(t_1)$$

$$2at_1 = 12$$

$$2(2)t_1 = 12$$

$$t_1 = 3$$

let normal meet parabola again at $Q(t_2)$

$$Q(t_2)$$

$$t_2 = -t_1 - \frac{2}{t_1} = -3 - \frac{2}{3} = -\frac{11}{3}$$

$$Q\left(2\left(-\frac{11}{3}\right)^2, 2(2)\left(-\frac{11}{3}\right)\right)$$

$$Q\left(\frac{242}{9}, -\frac{44}{3}\right)$$

$$PQ = \frac{80}{9}\sqrt{10}$$

Q24. Prove that, the normal to $y^2 = 12x$ at $(3, 6)$ meets the parabola again in $(27, -18)$ & circle on this normal chord as diameter is $x^2 + y^2 - 30x + 12y - 27 = 0$.

Ans. -

Sol. $y^2 = 12x$

let normal at P meet parabola

$$a = 3$$

again at $Q(t_2)$

$$P(3, 6) \quad t_2 = -t_1 - \frac{2}{E_1} = -1 - 2 = -3$$

$$t_1 = 1 \quad Q(27, -18)$$

circle with PQ as a diameter

$$(x - 2)(x - 27) + (y - 6)(y + 18) = 0$$

$$x^2 + y^2 - 30x + 12y - 27 = 0$$

Q25. Find the equation of the circle which passes through the focus of the parabola $x^2 = 4y$ & touches it at the point (6, 9).

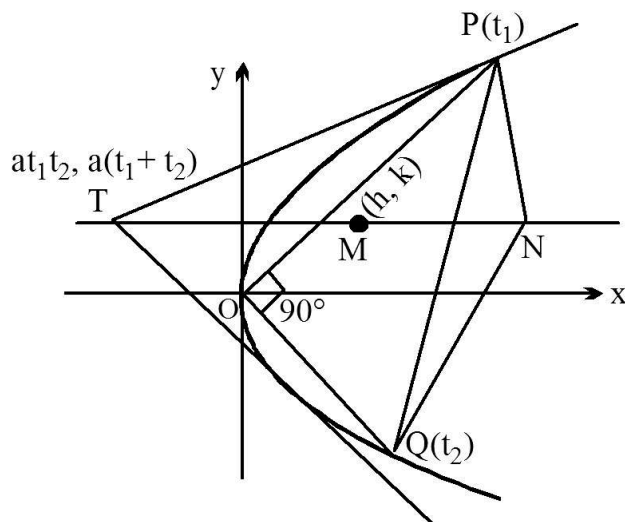
Ans. $x^2 + y^2 + 18x - 28y + 27 = 0$

Sol. -

Q26. PQ, a variable chord of the parabola $y^2 = 4x$ subtends a right angle at the vertex. The tangents at P and Q meet at T and the normals at those points meet at N. If the locus of the mid point of TN is a parabola, then find its latus rectum.

Ans. $25/2$

Sol.



$$\frac{2at_1}{at_1^2} \cdot \frac{2at_2}{at_2^2} = -1$$

$$t_1 t_2 = -4$$

$$\text{Now } T \equiv -4t_1 t_2, -2(t_1 + t_2)$$

$$\text{and } N \equiv (t_1^2 + t_2^2 + t_1 t_2 + 2), -2t_1 t_2(t_1 + t_2)$$

$$\text{Hence } 2h = a((t_1 + t_2)^2 + 2)$$

$$\& \quad 2k = a(t_1 + t_2)(1 - t_1 t_2)$$

$$2k = 5a(t_1 + t_2)$$

$$\therefore t_1 + t_2 = \frac{2k}{5a}$$

$$\therefore 2h = \left[\left(\frac{2k}{5a} \right)^2 + 2 \right]$$

$$2h = \left[\frac{4k^2 + 50a^2}{25a^2} \right]$$

$$50ah = 4k^2 + 50a^2$$

$$25ah = 2k^2 + 25a^2$$

$$25 \cdot 2(h - 2) = 2k^2 \quad \text{or} \quad y^2 = 25(x - 2)$$

Hence latus rectum = 25 **Ans.**

Q27. Let P be a point on the parabola $y^2 - 2y - 4x + 5 = 0$, such that the tangent on the parabola at P intersects the directrix at point Q. Let R be the point that divides the line segment PQ externally in the ratio $\frac{1}{2}$. Find the locus of R.

Ans. $2(y - 1)^2(x - 2) = (3x - 4)^2$

Sol. The given equation of parabola is

$$y^2 - 2y - 4x + 5 = 0 \quad \dots (1)$$

$$\Rightarrow (y - 1)^2 = 4(x - 1)$$

Any parametric point on this parabola is

$$P(t^2 + 1, 2t + 1)$$

Differentiating (1) w.r.t. x, we get

$$2y \frac{dy}{dx} - 2 \frac{dy}{dx} - 4 = 0$$

$$\Rightarrow \frac{dy}{dx} = \frac{2}{y - 1}$$

$\therefore$ Slope of tangent to (1) at pt.

$$P(t^2 + 1, 2t + 1) \text{ is } m = \frac{2}{2t} = \frac{1}{t}$$

$\therefore$ Equation of tangent at $P(t^2 + 1, 2t + 1)$ is

$$y - (2t + 1) = \frac{1}{t}(x - t^2 - 1)$$

$$\Rightarrow yt - 2t^2 - t = x - t^2 - 1$$

$$\Rightarrow x - yt + (t^2 + t - 1) = 0 \quad \dots (2)$$

Now direction of given parabola is

$$(x - 1) = -1 \Rightarrow x = 0$$

Tangent to (2) meets directrix at Q $\left(0, \frac{t^2 + t - 1}{t}\right)$

Let pt. R be (h, k)

ATQ R divides the line joining QP in the ratio

$\frac{1}{2} : 1$ i.e. $1 : 2$ externally.

$$\therefore (h, k) = \left[\frac{1(1+t^2) - 0}{-1}, \frac{t + 2t^2 - 2t^2 - 2t + 2}{-t} \right]$$

$$\Rightarrow h = -(1 + t^2) \text{ and } k = \frac{t - 2}{t}$$

$$\Rightarrow t^2 = -1 - h \text{ and } t = \frac{2}{1 - k}$$

Eliminating t we get $\left(\frac{2}{1 - k}\right)^2 = -1 - h$

$$\Rightarrow 4 = -(1 - k)^2 (1 - h)$$

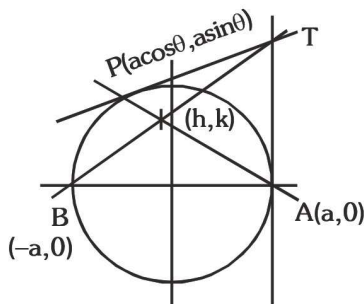
$$\Rightarrow (h - 1)(k - 1)^2 + 4 = 0$$

$\therefore$ locus of R(h, k) is, $(x - 1)(y - 1)^2 + 4 = 0$

- Q28. The tangent at any point P of a circle $x^2 + y^2 = a^2$ meets the tangent at a fixed point A (a, 0) in T and T is joined to B, the other end of the diameter through A, prove that the locus of the intersection of AP and BT is an ellipse whose eccentricity is $1/\sqrt{3}$.

Ans. -

Sol.



Equation of tangent at P

$$x \cos \theta + y \sin \theta = a$$

which meets $x = a$ at T

$$\therefore T \left(a, a \tan \frac{\theta}{2} \right)$$

$$\text{Equation of AP} \rightarrow y = -\cot\left(\frac{\theta}{2}\right)(x - a) \quad \dots(1)$$

$$\text{Equation of BT} \rightarrow y = \frac{\tan\left(\frac{\theta}{2}\right)}{2}(x + a) \quad \dots(2)$$

From (1) & (2)

$$y^2 - \frac{1}{2}(x^2 - a^2)$$

$$x^2 + 2y^2 = a^2$$

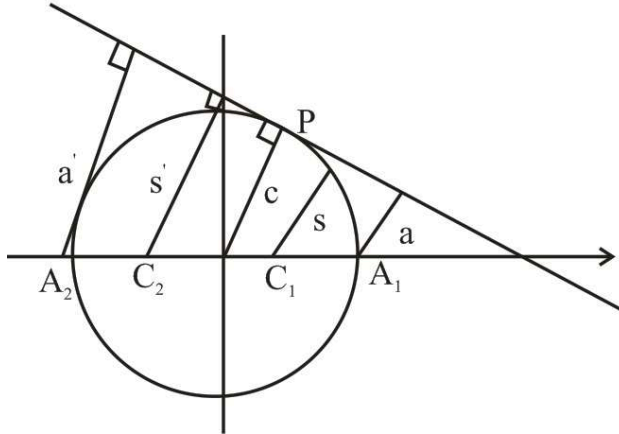
Q29. If s, s' are the length of the perpendicular on a tangent from the foci, a, a' are those from the vertices,

c is that from the centre and e is the eccentricity of the ellipse, $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$, then

prove that $\frac{ss' - c^2}{aa' - c^2} = e^2$

Ans. -

Sol.



equation of tangent

$$\frac{hx}{a^2} + \frac{ky}{b^2} = 1$$

$$s = \frac{\frac{h(ae)}{a^2} + 0 + 1}{\sqrt{\frac{h^2}{a^4} + \frac{k^2}{b^4}}} = \left| \frac{he + a}{\sqrt{\frac{h^2}{a^4} + \frac{k^2}{b^4}}} \right|$$

$$s' = \left| \frac{-he + a}{\sqrt{\frac{h^2}{a^4} + \frac{k^2}{b^4}}} \right|$$

$$a = \left| \frac{h + a}{\sqrt{\frac{h^2}{a^4} + \frac{k^2}{b^4}}} \right|$$

$$a' = \left| \frac{-h + a}{\sqrt{\frac{h^2}{a^4} + \frac{k^2}{b^4}}} \right|$$

$$c = \frac{1}{\sqrt{\frac{h^2}{a^4} + \frac{k^2}{b^4}}}$$

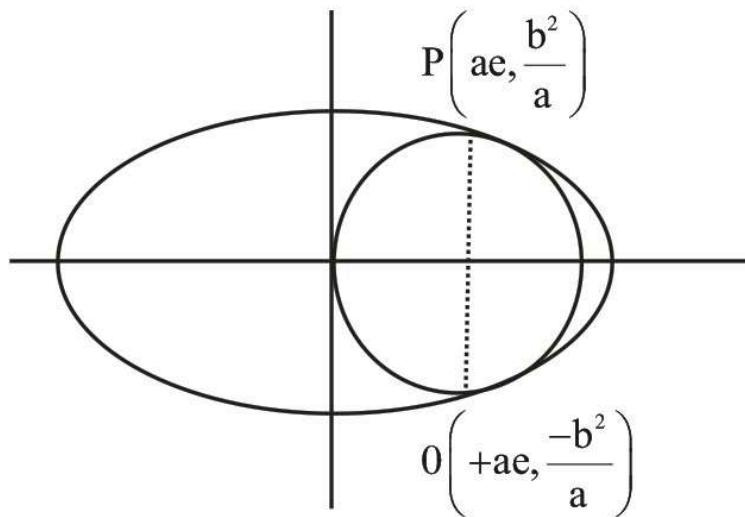
$$\frac{ss' - c^2}{aa' - c^2} = \frac{-h^2e^2 + a^2 - a^2}{-h^2 + a^2 - a^2} = e^2$$

Q30. Prove that the equation to the circle, having double contact with the ellipse

$\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ (with eccentricity e) at the ends of a latus rectum, is $x^2 + y^2 - 2ae^3x = a^2(1 - e^2 - e^4)$.

Ans. -

Sol.



tangent at P

$$\frac{xe}{a} + \frac{y}{a} = 1$$

$$xe + y - a = 0$$

$$Sp + \lambda L = 0$$

$$(x - ae)^2 + \left(y - \frac{b^2}{a}\right) + \lambda(xe + y - a) = 0$$

passing $(ae, -\frac{b^2}{a})$

$$0 + \left(-\frac{2b^2}{a}\right) + \lambda\left(ae^2 - \frac{b^2}{a} - a\right) = 0$$

$$\lambda = -\frac{-4b^4}{a^2\left(\frac{a^2e^2 - \frac{b^2}{a} - a}{a}\right)}$$

$$\lambda = \frac{-4b^4}{a(a^2 - b^2 - b^2 - a^2)}$$

$$\lambda = \frac{-4b^4}{-2ab^2} = \frac{2b^2}{a}$$

$$x^2 + a^2e^2 - 2xae + \frac{2b^2e}{a}x + \frac{2b^2}{a}y - 2b^2 + y^2\frac{b^4}{a^2} - \frac{2b^2}{a}y = 0$$

$$x^2 + y^2 + 2e\frac{(b^2 - a^2)}{a}x - 2b^2 + \frac{b^4}{a^2} = a^2e^2 = 0$$

$$x^2 + y^2 - 2ae^3x = 2b^2 - \frac{b^4}{a^2} - a^2e^2$$

Q31. Find the equations of the lines with equal intercepts on the axes & which touch the ellipse $\frac{x^2}{16} + \frac{y^2}{9} = 1$.

Ans. $x + y - 5 = 0$, $x + y + 5 = 0$

Sol. equal intetpy

$$m = -1$$

equation of tangent

$$y = mn \pm \sqrt{a^2m^2 + b^2}$$

$$y = -n \pm \sqrt{16 + 9}$$

$$x + y \pm 5 = 0$$

Q32. A tangent having slope $-\frac{4}{3}$ to the ellipse $\frac{x^2}{18} + \frac{y^2}{32} = 1$, intersects the axis of x & y in points A & B respectively. If O is the origin, find the area of triangle OAB.

Ans. 24 sq. units

Sol. $n = -\frac{4}{3}$

$$y = mn \pm \sqrt{a^2m^2 + b^2}$$

$$y = \frac{y}{x} \pm \sqrt{18 \times \frac{16}{9} + 32}$$

$$y = \frac{-4}{3}x + 8$$

$$4x + 3y = 24$$

$$A(6, 0) \quad B(0, 8)$$

$$\begin{aligned} \triangle AOB &= \frac{1}{2} \times 6 \times 8 \\ &= 24 \end{aligned}$$

Q33. Find the equation of the largest circle with centre (1, 0) that can be inscribed in the ellipse $x^2 + 4y^2 = 16$.

Ans. $(x - 1)^2 + y^2 = \frac{11}{3}$

Sol. -

Q34. Find the equation of the tangent to the hyperbola $x^2 - 4y^2 = 36$ which is perpendicular to the line $x - y + 4 = 0$.

Ans. $x + y \pm 3\sqrt{3} = 0$

Sol. Let m be the slope of the tangent. Since the tangent is perpendicular to the line $x - y = 0$

$$\therefore m \times 1 = -1 \Rightarrow m = -1$$

$$\text{Since } x^2 - 4y^2 = 36 \text{ or } \frac{x^2}{36} - \frac{y^2}{9} = 1$$

$$\text{Comparing this with } \frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$$

$$\therefore a^2 = 36 \text{ and } b^2 = 9$$

So the equation of tangents are $y = (-1)x + \sqrt{36x(-1)^2 - 9}$

$$y = -x \pm \sqrt{27} \Rightarrow x + y + 3\sqrt{3} = 0$$

Q35. Tangents are drawn to the hyperbola $3x^2 - 2y^2 = 25$ from the point $(0, 5/2)$. Find their equations.

Ans. $3x + 2y - 5 = 0$; $3x - 2y + 5 = 0$

Sol. $\frac{x^2}{\frac{25}{3}} - \frac{y^2}{\frac{25}{2}} = 1$

$$y = mn \pm \sqrt{a^2m^2 - b^2}$$

$$\frac{5}{2} = 0(m) \pm \sqrt{\frac{25}{3}m^2 - \frac{25}{2}}$$

$$\frac{25}{4} = \frac{25}{3}m^2 - \frac{25}{2}$$

$$\frac{75}{4} = \frac{25}{3}m^2$$

$$m = \pm \frac{3}{2}$$

$$y - \frac{5}{2} = \pm \frac{3}{2}(x - 0)$$

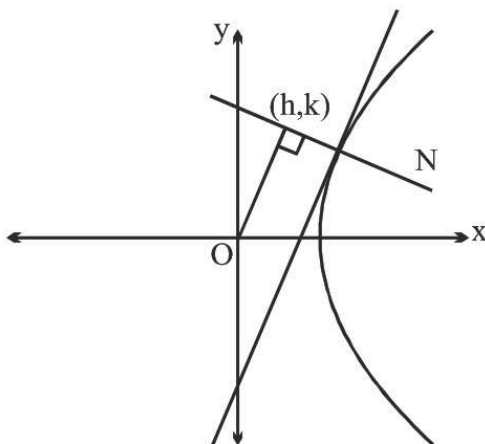
$$y \pm \frac{3}{2}x = \frac{5}{2}$$

$$2y \pm 3x = 5$$

Q36. Locus of the feet of the perpendicular from centre of the hyperbola $x^2 - 4y^2 = 4$ upon a variable normal to it has the equation, $(x^2 + y^2)^2 (4y^2 - x^2) = lx^2y^2$, find l.

Ans. 25

Sol.



Equation of N is

$$hx + ky = h^2 + k^2$$

Compare it with the normal at Q

$$\frac{ax}{\sec \theta} + \frac{by}{\tan \theta} = a^2 + b^2$$

$$\text{i.e., } \frac{2x}{\sec \theta} + \frac{y}{\tan \theta} = 5$$

$$\boxed{\frac{x^2}{4} - \frac{y^2}{1} = 1 \Rightarrow a = 2; b = 1}$$

$$\frac{h \cdot \sec \theta}{2} = k \tan \theta = \frac{h^2 + k^2}{5}$$

$$\sec \theta = \frac{2(h^2 + k^2)}{5h}$$

$$\tan \theta = \frac{h^2 + k^2}{5k}$$

$$\therefore \frac{4}{25} \frac{(h^2 + k^2)^2}{25h^2} - \frac{(h^2 + k^2)^2}{25k^2} = 1$$

$$4 \frac{(h^2 + k^2)^2}{20} \left(\frac{4}{h^2} - \frac{1}{k^2} \right) = 1 \Rightarrow k = 25 \Rightarrow A$$

Q37. Let P (a sec θ , b tan θ) and Q (a sec ϕ , b tan ϕ), where $\theta + \phi = \frac{\pi}{2}$, be two points on the hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$. If (h, k) is the point of intersection of the normals at P & Q, then find k.

Ans. $-\left(\frac{a^2 + b^2}{b}\right)$

Sol. Equation of normal at P.

$$N_1 : \frac{a^2 x}{x_1} + \frac{b^2 y}{y_1} = a^2 + b^2$$

$$ax \cos \theta + by \cot \theta = a^2 + b^2 \quad \dots (1)$$

equation of normal at Q (a sec ϕ , b tan ϕ)

$$Q(a \operatorname{cosec} \phi, b \cot \phi)$$

$$ax \sin \theta + by \tan \theta = a^2 + b^2 \quad \dots (2)$$

$$(1)x \sin \theta - (2)x \cos \theta$$

$$\text{by } (\cos \theta - \sin \theta) = (a^2 + b^2)(\sin \theta - \cos \theta)$$

$$y = \frac{-(a^2 + b^2)}{b}$$

Point of intersection is (h, k)

$$k = -\frac{(a^2 + b^2)}{b}$$

Q38.

Ans. -

Sol. -

Q39. Three normals to $y^2 = 4x$ pass through the point (15, 12). Show that if one of the normals is given by $y = x - 3$ & find the equations of the others.

Ans. $y = -4x + 72$, $y = 3x - 33$

Sol. $y = mn - 2am - am^3$

$a = 1$, passing (15, 12)

$$12 = 15m - 2m - m^3$$

$$m^3 - 13m + 12 = 0$$

$$m^3 - 13m + 12 = 0$$

$$(m - 1)(m - 3)(m + 4) = 0$$

$$m = 1, m = 3, m = -4$$

$$y = 3x - 6 - 27$$

$$y = -4x + 8 + 64$$

$$y = 3x - 33$$

$$y = -4x + 72$$

Q40. From the point P(h,k) three normals are drawn to the parabola $x^2 = 8y$ and m_1, m_2 and m_3 are the slopes of three normals

(a) Find the algebraic sum of the slopes of these three normals.

(b) If two of the three normals are at right angles then the locus of point P is a conic, find the latus rectum of conic.

(c) If the two normals from P are such that they make complementary angles with the axis then the locus of point P is a conic, find a directrix of conic.

Ans. (a) $\frac{k-4}{h}$; (b) 2; (c) $2y - 3 = 0$

Sol. $P(h, k)$

$$x^2 = 8y$$

passing (h, k)

$$kmm^2 = hm^3 + 2am^2 + a$$

$$hm^3 + (2a - k)m^2 + a = 0$$

$$hm^3 + (4 - k)m^2 + 2 = 0$$

$$(a) m_1 = m_2 + m_3 = \frac{(4 - k)}{h}$$

$$(b) m_1 = m_2 = -1, \quad m_1 m_2 m_3 = \frac{-3}{h} \Rightarrow m_3 = \frac{2}{h}$$

$$h\left(\frac{2}{h}\right)^3 + (4-k)\left(\frac{2}{h}\right)^2 + 2 = 0$$

$$\frac{B}{h^2} + \frac{4}{h^2}(4-k) + 2 = 0$$

$$8 + 16 - 4k + 2h^2 - 12$$

$$x^2 = 2y - 12$$

$$x^2 = 2(y - 6)$$

$$l_{LR} = 2$$

$$(C) \ m_1 m_2 = 1$$

$$m_1 m_2 m_3 = \frac{-2}{h}$$

$$m_3 = \frac{-2}{h}$$

$$h\left(\frac{-2}{h}\right)^3 + (4-k)\left(\frac{-2}{h}\right)^2 + 2 = 0$$

$$-\frac{8}{2} + (4-k)\frac{4}{h^2} + 2 = 0$$

$$-8 + 16 - 4k + 2h^2 = 0$$

$$8 - 4k + 2h^2 = 0$$

$$x^2 = 2(y - 2)$$

$$\text{Directrix} \quad y - 3 = -\frac{1}{2} \quad (y = -a)$$

$$2y - 4 = -1$$

$$2y - 3 = 0$$

Q41. Prove that the two parabolas $y^2 = 4ax$ & $y^2 = 4c(x - b)$ cannot have a common normal, other than the axis, unless $\frac{b}{(a - c)} > 2$.

Ans. -

$$\text{Sol.} \quad y^2 = 4an, \quad y^2 = 4c(x - b)$$

Let slope of common normal is m

$$y = mn - 2am - am^3 \quad \dots\dots (1) \quad \left[\text{for } y^2 = 4an \right]$$

$$y = m(x - b) - 2cm - cm^3 \quad \left[\text{for } y^2 = 4c(n - b) \right]$$

$$y = mn - mb - 2cm - cm^3 \quad \dots\dots(2)$$

For common normal

$$-2am - am^3 = -mb - 2mc - cm^3$$

$$2a + m^2 = b + 3c + cm^2$$

$$(2a - b - 2c) = (c - a)m^2$$

$$m^2 = \frac{(2(a - c) - b)}{-(a - c)}$$

$$\therefore m^2 > 0 \quad [m \neq 0]$$

$$-\frac{2(a \cdot c)}{a - c} + \frac{b}{a - c} > 0$$

$$\boxed{\frac{b}{(a \cdot c)} > 2}$$

Q42. Find the condition on 'a' & 'b' so that the two tangents drawn to the parabola $y^2 = 4ax$ from a point are normals to the parabola $x^2 = 4by$.

Ans. $a^2 > 8b^2$

Sol. Equation of tangent of $y^2 = 4ax$ in slope form at (x_1, y_1) is

$$y_1 = mx_1 + \frac{a}{m} \quad \dots (1)$$

equation of normal at $(2bt_1, bt_1^2)$ on $x^2 = 4by$

$$x + t_1y = 2bt_1 + bt_1^3$$

It passes through (x_1, y_1)

$$\therefore x_1 + t_1y_1 = 2bt_1 + bt_1^3 \quad \dots (2)$$

(1) & (2) are same equation so compare

$$\frac{1}{t_1} = -\frac{m}{1} = \frac{a}{m(2bt_1 + bt_1^3)}$$

$$t_1m = -1$$

$$-m^2t_1(2b + bt_1^2) = a$$

$$\Rightarrow m(2b + bt_1^2) = a \quad \dots (3)$$

Put $m = -\frac{1}{t_1}$ in equation (3)

$$2b + bt_1^2 = -at_1$$

$$bt_1^2 + at_1 + 2b = 0$$

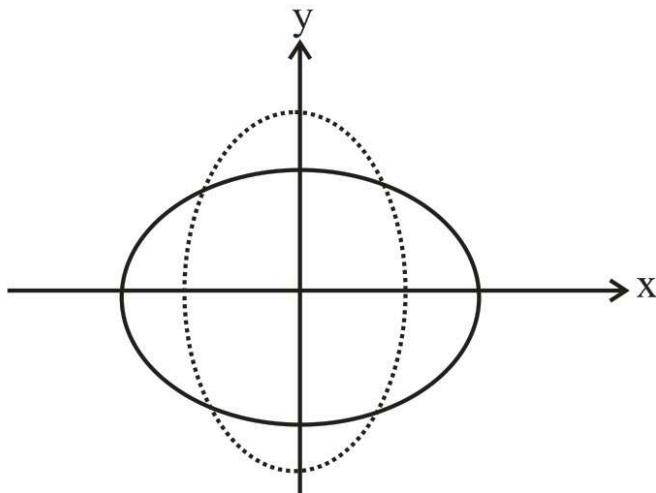
t_1 will be real

$$a^2 > 8b^2$$

Q43. Find the equation of the common tangent in 1st quadrant to the circle $x^2 + y^2 = 16$ and the ellipse $\frac{x^2}{25} + \frac{y^2}{4} = 1$. Also find the length of the intercept of the tangent between the coordinate axes.

Ans. $AB = \frac{14}{\sqrt{3}}$

Sol.



Let slope of common tangent is m

$$y = mn \pm \sqrt{am^2 + a^2} \quad (\text{for circle})$$

$$a = 4$$

$$y = mn \pm \sqrt{16m^2 + 16} \quad \dots\dots (1)$$

For ellipse

$$y = mn \pm \sqrt{16m^2 + 16} \quad \dots\dots (2)$$

For common tangent

$$\sqrt{16m^2 + 16} = \sqrt{25m^2 + 4}$$

$$16m^2 + 16 = 25m^2 + 4$$

$$9m^2 = 12$$

$$m \pm \frac{2}{\sqrt{3}}$$

$$\text{for 1}^{\text{st}} \text{ quadrant } m = \frac{-2}{\sqrt{3}}$$

$$y = -\frac{2}{\sqrt{3}} = 2\frac{28}{\sqrt{3}}$$

$$\sqrt{3}y + 2x = 4\sqrt{7}$$

cut x-axis at $A(2\sqrt{7}, 0)$

cut y-axis at $B\left(0, \frac{4\sqrt{7}}{\sqrt{3}}\right)$

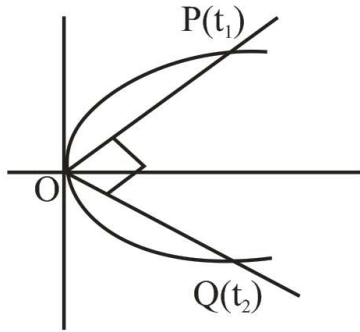
$$\begin{aligned} AB &= \sqrt{4(7) + 16\left(\frac{7}{3}\right)} \\ &= 2\sqrt{7 + \frac{28}{3}} \end{aligned}$$

$$= \frac{14}{\sqrt{3}}$$

Q44. Through the vertex O of a parabola $y^2 = 4x$, chords OP & OQ are drawn at right angles to one another. Show that for all positions of P, PQ cuts the axis of the parabola at a fixed point. Also find the locus of the middle point of PQ.

Ans. $(4, 0)$; $y^2 = 2a(x - 4a)$

Sol.



$$OP \perp OQ$$

$$t_1 t_2 = -4 \quad \Rightarrow \quad t_2 = \frac{-4}{t_1}$$

$$P(t_1^2, 2t_1) \quad Q(t_2^2, 2t_2)$$

$$Q\left(\frac{16}{t_1^2}, \frac{-8}{t_1}\right)$$

Equation of PQ

$$y - 2t_1 = \frac{\left(-\frac{8}{t_1} - 2t_1\right)}{\left(\frac{16}{t_1^2} - t_1^2\right)}(x - t_1^2)$$

$$y - 2t_1 = \frac{-2(4 + t_1^2)}{t_1 \frac{(16 - t_1^4)}{t_1^2}}(x - t_1^2)$$

$$y - 2t_1 = -2t_1 \frac{(4 + t_1^2)}{(4 - t_1^2)(4 + t_1^2)}(x - t_1^2)$$

$$(y - 2t_1)(4 - t_1^2) = -2xt_1 + 2t_1^3$$

$$4y - 8t_1 - t_1^3 y + 2t_1(x - 4) = 0$$

$$y + \frac{2t_1}{(4 - t_1^3)}(x - 4) = 0$$

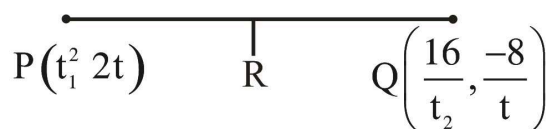
$$L_1 + \lambda L_2 = 0$$

$$L_1 = 0, \quad L_2 = 0$$

$$y = 0 \quad x = 4$$

passing fixed point on x - axis $(4, 0)$

let mid point of PQ is R (h, k)



$$h = \frac{t^2 + \frac{16}{t^2}}{2}$$

$$k = \frac{2t - \frac{8}{t}}{2}$$

$$2h = t^2 + \frac{16}{t^2} \dots (1)$$

$$k = t - \frac{4}{t}$$

$$k^2 = t^2 + \frac{16}{t^2} - 8 \dots (2)$$

From (1) & (2)

$$2h = k^2 + 8$$

$$y^2 = 2x - 8$$

$$y^2 = 2(x - 4)$$

Q45. From the point $(-1, 2)$ tangent lines are drawn to the parabola $y^2 = 4x$. Find the equation of the chord of contact. Also find the area of the triangle formed by the chord of contact & the tangents.

Ans. $x - y = 1$; $8\sqrt{2}$ sq. units

Sol.

Equation of AB, C.O.C from p

$$T = 0$$

$$yy_1 = 2(x + x_1)$$

$$2y = 2(x - 1)$$

$$y = x - 1$$

$$x - y - 1 = 0$$

$$h = \left| \frac{1 + 2 + 1}{\sqrt{2}} \right| = 2\sqrt{2}$$

$$AB = \frac{4}{m^2} \sqrt{a(1 + m^2)(a - mc)}$$

$$a = 1, m = 1, c = -1$$

$$= \frac{4}{(1)^2} \sqrt{1(1 + 1)(1 + 1)}$$

$$= 4\sqrt{4} = 8$$

$$\triangle PAB = \frac{1}{2} AB(h)$$

$$= \frac{1}{2} 8 \times 2\sqrt{2} = 8\sqrt{2}$$

Q46. Consider the family of circles, $x^2 + y^2 = r^2$, $2 < r < 5$. If in the first quadrant, the common tangent to a circle of the family and the ellipse $4x^2 + 25y^2 = 100$ meets the co-ordinate axes at A & B, then find the equation of the locus of the mid-point of AB.

Ans. $25y^2 + 4x^2 = 4x^2y^2$

Sol. $x^2 + y^2 = r^2$, $\frac{x^2}{25} + \frac{y^2}{4} = 1$

let slop of common tangent is m

C.O.T for circle = C.O.T for ellipse

$$r\sqrt{1+m^2} = r\sqrt{25m^2+4}$$

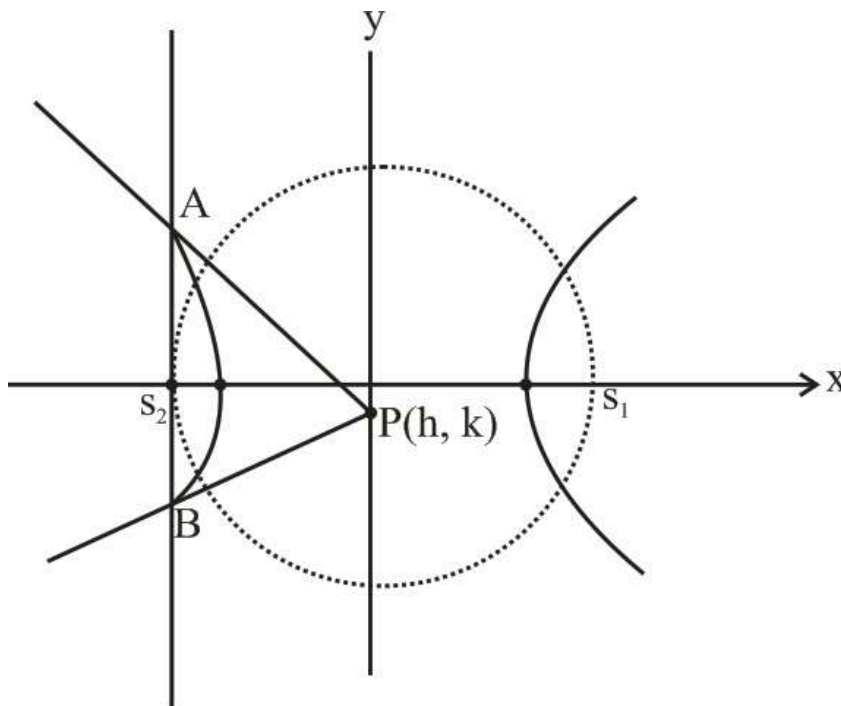
$$r^2 + r^2m^2 = \sqrt{25m^2+4}$$

$$m^2 = \frac{4-r^2}{r^2-25}$$

Q47. Chords of the hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ are tangents to the circle drawn on the line joining the foci as diameter. Find the locus of the point of intersection of tangents at the extremities of the chords.

Ans. $\frac{x^2}{a^4} + \frac{y^2}{b^4} = \frac{1}{a^2+b^2}$

Sol.



AB is COC of hyperbola w.r.t

$$T = 0$$

$$\frac{hx}{a^2} - \frac{ky}{b^2} = 1$$

equation of circle $\Rightarrow x^2 + y^2 = a^2 + b^2$

$$x^2 + y^2 = a^2 + b^2$$

Line (1) is tangent to circle

$$\left| \frac{0 - 0 - 1}{\sqrt{\frac{h^2}{a^4} + \frac{k^2}{b^4}}} \right| = \sqrt{a^2 + b^2}$$

$$\frac{x^2}{a^4} + \frac{y^2}{b^4} = \frac{1}{a^2 + b^2}$$

Q48. Tangents are drawn from any point on the hyperbola $\frac{x^2}{9} - \frac{y^2}{4} = 1$ to the circle $x^2 + y^2 = 9$. Find the locus of midpoint of the chord of contact.

Ans. $\frac{x^2}{9} - \frac{y^2}{4} = \left(\frac{x^2 + y^2}{9} \right)^2$

Sol. Let point on hyperbola $(3 \sec \theta + 2 \tan \theta)$

$$T = 0$$

$$x(3 \cos \theta) + y(2 \tan \theta) = 9 \quad \dots (1)$$

Let mid point of C.O.C is R(h, k)

then equation of chord whose mid point is given

$$T = S_1$$

$$hx + ky = h^2 + k^2 \quad \dots (2)$$

$$\frac{3 \sec \theta}{h} = \frac{2 \tan \theta}{k} = \frac{9}{h^2 + k^2}$$

$$\sec \theta = \frac{9h}{3(h^2 + k^2)} \quad \tan \theta = \frac{9k}{2(h^2 + k^2)}$$

$$\sec^2 \theta - \tan^2 \theta = 1$$

$$\frac{81x^2}{9(x^2 + y^2)^2} = \frac{81y^2}{4(x^2 + y^2)^2} = 1$$

Q49. Find the number of integral values of parameter 'a' for which three chords of the ellipse $\frac{x^2}{2a^2} + \frac{y^2}{a^2} = 1$ (other than its diameter) passing through the point $p\left(11a, -\frac{a^2}{4}\right)$ are bisected by the parabola $y^2 = 4ax$.

Ans. 2

Sol. Any point on the parabola $y^2 = 4ax$ is $(at^2, 2at)$. Equation of chord of the ellipse

$$\frac{x^2}{2a^2} + \frac{y^2}{a^2} = 1, \text{ whose mid-point is } (at^2, 2at) \text{ is } \frac{xat^2}{2a^2} + \frac{y2at}{a^2} = \frac{a^2t^4}{2a^2} + \frac{4a^2t^2}{a^2}$$

$$\Rightarrow tx + 4y = at^3 + 8at \quad (\because t \neq 0)$$

As it passes through $\left(11a, -\frac{a^2}{4}\right)$,

$$\Rightarrow 11at - 4\left(\frac{a^2}{4}\right)at^3 + 8at \Rightarrow at^3 - 3at = a^2 = 0$$

$$\Rightarrow t^3 - 3t + a = 0 \quad (a \neq 0)$$

Now, three chords of the ellipse will be bisected by the parabola if the equation (1) has three real and distinct roots.

Now, three chords of the ellipse will be bisected by the parabola if the equation (1) has three real and distinct roots.

$$\text{Let } f(t) = t^3 - 3t + a$$

$$f'(t) = 3t^2 - 3 = 0 \Rightarrow t = \pm 1$$

$$\text{So, } f(1)f(-1) < 0$$

$$\Rightarrow a \in (-2, 2)$$

$$\text{But } a \in 0, \text{ so } a \in (-2, 0) \cup (0, 2)$$

$\therefore$ Number of integral values of 'a' = 2.]

Q50. A variable circle passes through the point A (2, 1) and touches the x-axis. Locus of the other end of the diameter through A is a parabola.

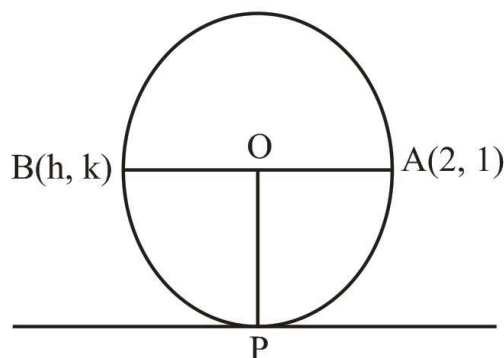
(a) Find the length of the latus rectum of the parabola.

(b) Find the coordinates of the foot of the directrix of the parabola.

(c) The two tangents and two normals at the extremities of the latus rectum of the parabola constitutes a quadrilateral. Find area of quadrilateral.

Ans. (a) 4, (b) (2, -1), (c) 8 sq. units

Sol.



$$OP = \frac{AB}{2}$$

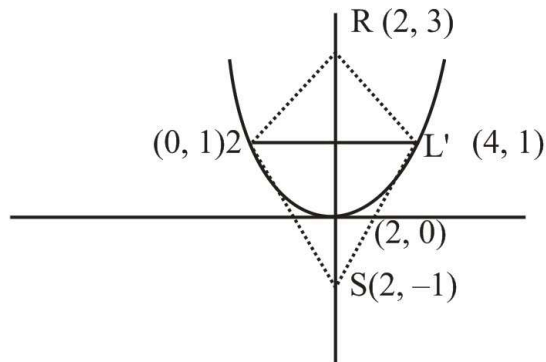
$$O\left(\frac{2+h}{2}, \frac{k+1}{2}\right)$$

$$\left(\frac{k+1}{2}\right)^2 = \frac{1}{4}\left((h-2)^2 + (k-1)^2\right)$$

$$(k+1)^2 = (h-2)^2 + (k-1)^2$$

$$k^2 + 2k + 1 = h^2 - 4h + 4 + k^2 - 2k + 1$$

$$(x-2)^2 = 4y$$



(a) $L'R = 4$

(b) $Z(2, -1)$

(c) $\frac{1}{2}(4)(4) = 8$

Q51. Show that the circle through three points the normals at which to the parabola $y^2 = 4ax$ are concurrent at the point (h, k) is $2(x^2 + y^2) - 2(h + 2a)x - ky = 0$.

Ans. -

Sol. Equation of normal

$$y = mn - 2am - am^3$$

passing (h, k)

$$am^3 + (2a - h)m + k = 0 \quad \begin{matrix} \swarrow m_1 \\ = m_2 \\ \searrow m_3 \end{matrix} \quad \dots (1)$$

let co-normal point are $(am_1^2, -2am_1)$

$$m_1 + m_2 + m_3 = 0$$

let equation circle is

$$x^2 + y^2 + 2gx + 2fy + c = 0$$

passing through $(am_1^2, -2am_1)$

$$a^2m^4 + 4a^2m^2 + 2agm^2 - 4afm + c = 0$$

$$a^2m^4 + (4a^2 + 2ag)m^2 - 4afm + c = 0 \quad \begin{matrix} \swarrow m_1 \\ = m_2 \\ \searrow m_3 \\ m_4 \end{matrix}$$

$$m_1 + m_2 + m_3 + m_4 = 0$$

$$m_4 = 0 \quad [\because m_1 + m_2 + m_3 = 0]$$

$$c = 0$$

$$m(a^2m^3 + (4a^2 + 2ag)m - 4af) = 0$$

$$a^2m^3 + (4a^2 + 2ag)m - haf = 0 \quad \dots (2)$$

equation (1) & (2) are same

$$\frac{a}{a^2} = \frac{2a - h}{4a^2 + 2ag} = \frac{k}{-4af}$$

$$f = \frac{-k}{4}, \quad 2a - h = 4a + 2g$$

$$2g = -(2a + h)$$

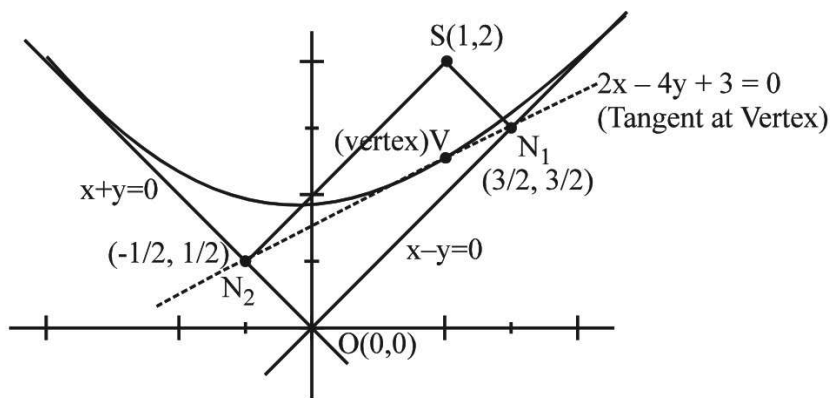
$$n^2 = y^2 - (2a + h) - \frac{k}{2}y = 0$$

Q52. Let $L_1 : x + y = 0$ and $L_2 : x - y = 0$ are tangent to a parabola whose focus is $S(1, 2)$.

If the length of latus-rectum of the parabola can be expressed as $\frac{m}{\sqrt{n}}$ (where m and n are coprime) then find the value of $(m + n)$.

Ans. 11

Sol.



Feet of the perpendicular (N_1 and N_2) from focus upon any tangent to parabola lies on the tangent line at the vertex.

Now equation of SN_1 is $x + y = 1$ passing through $(1, 2) \Rightarrow 1 = 3$

Equation of SN_1 is $x + y = 3$

Solving $x + y = 3$ and $y = x$, we get $N_1 \equiv \left(\frac{3}{2}, \frac{3}{2}\right)$

|||y equation of SN_2 is $x - y = 1$ passing through $(1, 2) \Rightarrow \lambda = -1$

Equation of SN_2 is $y - x = 1$

Solving $y - x = 1$ and $y = -x$, we get $N_2 \equiv \frac{|2 - 8 + 3|}{\sqrt{20}} = \frac{3}{2\sqrt{5}} = \frac{1}{4} \times \text{latus rectum}$

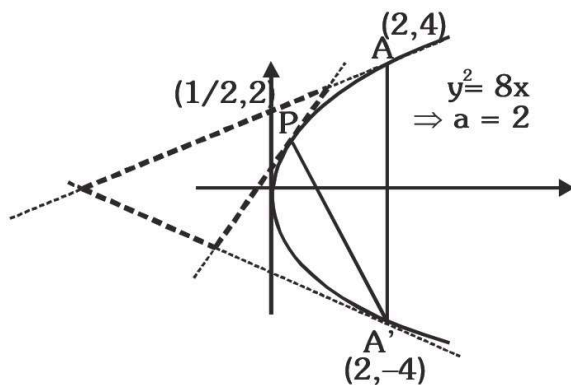
and hence length of latus rectum = $\frac{6}{\sqrt{5}} = \frac{m}{\sqrt{n}}$

Hence $m + n = 6 + 5 = 11$ **Ans.**

- Q53. Consider the parabola $y^2 = 8x$. Let Δ_1 be the area of the triangle formed by the end points of its latus rectum and the point $P\left(\frac{1}{2}, 2\right)$ on the parabola, and Δ_2 be the area of the triangle formed by drawing tangents at P and at the end points of the latus rectum. Then $\frac{\Delta_1}{\Delta_2}$ is

Ans. 2

Sol.



$$\Delta_1 = \text{area of } \triangle PAA' = \frac{1}{2} \cdot 8 \cdot \frac{3}{2} = 6$$

$$\Delta_2 = \frac{1}{2}(\Delta_1)$$

(Using property : Area of triangle formed by tangents is always half of original triangle)

$$\Rightarrow \frac{\Delta_1}{\Delta_2} = 2$$

- Q54. Let d be the perpendicular distance from the centre of the ellipse to the tangent drawn at a point P on the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$. If F_1 & F_2 are the two foci of the ellipse, then show that $(PF_1 - PF_2)^2 = 4a^2 \left[1 - \frac{b^2}{d^2} \right]$.

Ans. -

Sol. Equation of tangent at point P ($a \cos \theta$, $b \sin \theta$)

$$\text{on } \frac{x^2}{a^2} + \frac{y^2}{b^2} = 1 \text{ is } \frac{x}{a} \cos \theta + \frac{y}{b} \sin \theta = 1$$

$$\text{foci } F_1 \equiv (ae, 0), F_2 \equiv (-ae, 0)$$

$$\text{and } d = \frac{1}{\sqrt{\frac{\cos^2 \theta}{a^2} + \frac{\sin^2 \theta}{b^2}}} = \frac{ab}{\sqrt{a^2 \sin^2 \theta + b^2 \cos^2 \theta}}$$

$$\begin{aligned}
& \text{Now } 4a^2 \left(1 - \frac{b^2}{d^2}\right) \\
& 4\cos^2\theta (a^2 - b^2) \\
& = 4a^2 e^2 \cos^2\theta = (2ae \cos\theta)^2 \\
& = [(a - ae \cos\theta) - (a + ae \cos\theta)]^2 \\
& = (PF_1 - PF_2)^2
\end{aligned}$$

Q55. An ellipse has foci at $F_1(9, 20)$ and $F_2(49, 55)$ in the xy -plane and is tangent to the x -axis. Find the length of its major axis.

Ans. 85

Sol. $PS_1 + PS_2 = 2a$

$$S_1 S_2^1 = 2a$$

$$S_1 S_2^1 = 2a$$

$$(49 - 9)^2 + (55 - 20)^2 = (2a)^2$$

$$2a = 85$$

Q56. If C is the centre of a hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$, S, S' its foci and P a point on it. Prove that $SP \cdot S'P = CP^2 - a^2 + b^2$.

Ans. -

Sol. Let $P(a \sec \theta, b \tan \theta)$

$$SP = epm = e \left(a \sec \theta - \frac{a}{e} \right) = (ae \sec \theta - a)$$

$$SP' = epm' = e \left(a \sec \theta + \frac{a}{e} \right) = (ae \sec \theta + a)$$

$$SP \cdot SP' = (ae \sec \theta - a)(ae \sec \theta + a)$$

$$= a^2 e^2 \sec^2 \theta - a^2$$

$$= (a^2 + b^2) \sec^2 \theta - a^2$$

$$= b^2 \sec^2 \theta = a^2 \tan^2 \theta$$

$$CP^2 = a^2 \sec^2 \theta + b^2 \tan^2 \theta$$

$$CP^2 - a^2 + b^2 = a^2 \sec^2 \theta + b^2 \tan^2 \theta - a^2 + b^2$$

$$= a^2 \tan^2 \theta = b^2 \sec^2 \theta$$

Q57. A conic C satisfies the differential equation, $(1 + y^2) dx - xy dy = 0$ and passes through the point $(1, 0)$. An ellipse E which is confocal with C having its eccentricity equal to $\sqrt{2/3}$.

- (a) Find the length of the latus rectum of the conic C
 (b) Find the equation of the ellipse E.
 (c) Find the locus of the point of intersection of the perpendicular tangents to the ellipse E.

Ans. (a) 2; (b) $\frac{x^2}{3} + \frac{y^2}{1} = 1$; (c) $x^2 + y^2 = 4$

Sol. $(1 + y^2)dx = xydy$

$$\frac{dx}{x} = \frac{ydy}{1 + y^2}$$

$$\ln x = \frac{1}{2} \ln(1 + y^2) + \ln c$$

$$\frac{x^2}{1 + y^2} = C$$

passing (1, 0)

$$x^2 = 1 + y^2$$

$$C : x^2 - y^2 = 1$$

(a) $l_{LR} = 2$

(b) $t : \frac{x^2}{A^2} + \frac{y^2}{B^2} = 1$

can focal with 'c'

$$Ae' = ae$$

$$A. \sqrt{\frac{2}{3}} = \sqrt{2}$$

$$A = \sqrt{3}, \quad B = 1$$

$$\frac{x^2}{3} + \frac{y^2}{1} = 1$$

(c) locus of P.O.I of perpendicular tangents is

Director circle of ellipse

$$x^2 + y^2 = 3 + 1$$

$$x^2 + y^2 = 4$$

- Q58. Let 'p' be the perpendicular distance from the centre C of the hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ to the tangent drawn at a point R on the hyperbola. If S & S' are the two foci of the hyperbola, then show that $(RS + RS')^2 = 4a^2 \left(1 + \frac{b^2}{p^2}\right)$.

Ans. -

Sol. Let point on hyperbola $R(a \sec \theta, b \tan \theta)$

tangent at 'R'

$$\frac{x}{a} \sec \theta - \frac{y}{b} \tan \theta = 1$$

$$p = \frac{1}{\sqrt{\frac{\sec^2 \theta}{a^2} + \frac{\tan^2 \theta}{b^2}}}$$

$$\frac{b^2}{p^2} = \frac{b^2 \sec^2 \theta}{a^2} + \tan^2 \theta$$

$$1 + \frac{b^2}{p^2} = \frac{b^2}{a^2} \sec^2 \theta + \sec^2 \theta$$

$$4a^2 \left(1 + \frac{b^2}{a^2}\right) = 4 \sec^2 \theta \left(\frac{a^2 + b^2}{a^2}\right) a^2$$

$$= 4 \sec^2 \theta (a^2 + b^2)$$

$$\text{LHS } (RS + RS')^2 = \left(e(a \sec \theta - \frac{a}{e} + a \sec \theta + \frac{a}{e})\right)^2$$

$$= 4a^2 \sec^2 \theta \cdot e^2$$

$$= 4 \sec^2 \theta \cdot (a^2 + b^2) = \text{RHS}$$

Paragraph

Tangents are drawn from the point P(3, 4) to the ellipse $\frac{x^2}{9} + \frac{y^2}{4} = 1$ touching the ellipse at points A and B.

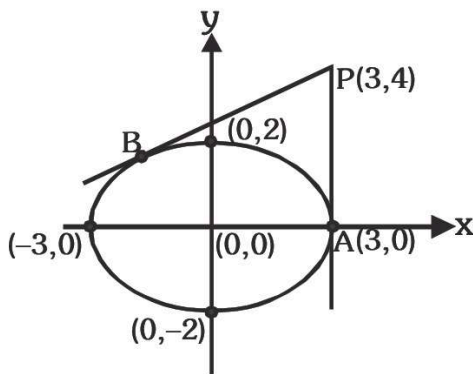
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Q59. The coordinates of A and B are

- (A) (3, 0) and (0, 2) (B) $\left(-\frac{8}{5}, \frac{2\sqrt{161}}{15}\right)$ and $\left(-\frac{9}{5}, \frac{8}{5}\right)$
- (C) $\left(-\frac{8}{5}, \frac{2\sqrt{161}}{15}\right)$ and (0, 2) (D) (3, 0) and $\left(-\frac{9}{5}, \frac{8}{5}\right)$

Ans. D

Sol. As shown in figure, one of the point of contact is (3,0)



Let equation of other tangent,

$$y = mx + \sqrt{9m^2 + 4} \text{ as } c > 0$$

It passes through (3,4), so

$$4 = 3m + \sqrt{9m^2 + 4}$$

$$(4 - 3m)^2 = 9m^2 + 4$$

$$\text{Solving, } m = \frac{1}{2}$$

As we know that point of contact for the tangent

$$\text{given by } \left(-\frac{a^2m}{\sqrt{a^2m^2 + b^2}}, \frac{b^2}{\sqrt{a^2m^2 + b^2}} \right)$$

$$\therefore \text{Point of contact is } \left(-\frac{9}{5}, \frac{8}{5} \right)$$

Paragraph

Tangents are drawn from the point P(3, 4) to the ellipse $\frac{x^2}{9} + \frac{y^2}{4} = 1$ touching the ellipse at points A and B.

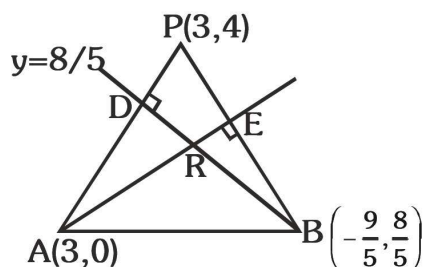
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Q60. The orthocenter of the triangle PAB is

$$(A) \left(5, \frac{8}{7} \right) \quad (B) \left(\frac{7}{5}, \frac{25}{8} \right) \quad (C) \left(\frac{11}{5}, \frac{8}{5} \right) \quad (D) \left(\frac{8}{25}, \frac{7}{5} \right)$$

Ans. C

Sol.



$$\text{Equation of line BD : } y = \frac{8}{5}$$

$$\text{Equation of line AE : } 2x + y = 6$$

Now orthocentre R of DPAB will be intersection of line BD and line AE.

$$\text{Solving for R, we get } R \equiv \left(\frac{11}{5}, \frac{8}{5} \right)$$

Q61. The equation of the locus of the point whose distances from the point P and the line AB are equal, is

$$(A) 9x^2 + y^2 - 6xy - 54x - 62y + 241 = 0 \quad (B) x^2 + 9y^2 + 6xy - 54x + 62y - 241 = 0$$

(C) $9x^2 + 9y^2 - 6xy - 54x - 62y - 241 = 0$ (D) $x^2 + y^2 - 2xy + 27x + 31y - 120 = 0$

Ans. A

Sol. Equation of line AB is $x + 3y = 3$

Now let the point be (h, k)

According to question,

$$\left| \frac{h + 3k - 3}{\sqrt{1^2 + 3^2}} \right| = \sqrt{(h - 3)^2 + (4 - k)^2}$$

After solving, we get

$$9x^2 + y^2 - 6xy - 54x - 62y + 241 = 0$$

Paragraph

The circle $x^2 + y^2 - 8x = 0$ and hyperbola $\frac{x^2}{9} - \frac{y^2}{4} = 1$ intersect at the points A and B.

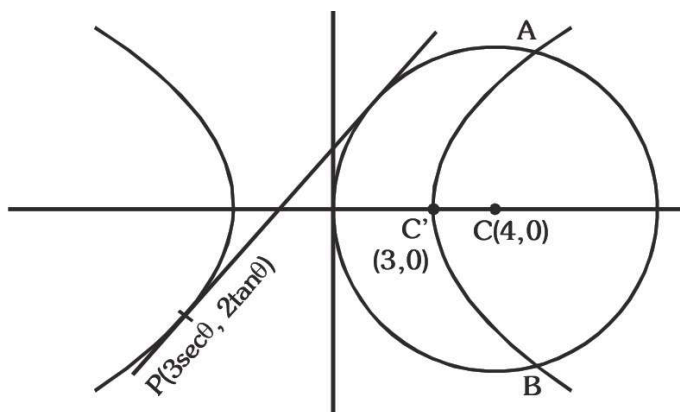
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Q62. Equation of a common tangent with positive slope to the circle as well as to the hyperbola is

(A) $2x - \sqrt{5}y - 20 = 0$ (B) $2x - \sqrt{5}y + 4 = 0$ (C) $3x - 4y + 8 = 0$ (D) $4x - 3y + 4 = 0$

Ans. B

Sol.



Equation of tangent $\frac{x \sec \theta}{3} - \frac{y \tan \theta}{2} = 1$

$$|p| = r$$

$$\frac{\left| \frac{4}{3} \sec \theta - 1 \right|}{\sqrt{\frac{\sec^2 \theta}{9} + \frac{\tan^2 \theta}{4}}} = 4$$

$$\Rightarrow \frac{16}{9} \sec^2 \theta + 1 - \frac{8}{3} \sec \theta = 16 \left(\frac{4 \sec^2 \theta + 9 \tan^2 \theta}{4 \times 9} \right)$$

$$16 \sec^2 \theta + 9 - 24 \sec \theta = 52 \sec^2 \theta - 36$$

$$\Rightarrow 36 \sec^2 \theta + 24 \sec \theta - 45 = 0$$

$$\Rightarrow 12 \sec^2 \theta + 8 \sec \theta - 15 = 0$$

$$\Rightarrow 12\sec^2\theta + 18\sec\theta - 10\sec\theta - 15 = 0$$

$$\Rightarrow (6\sec\theta - 5)(2\sec\theta + 3) = 0$$

$$\sec\theta = \frac{5}{6} \text{ (not possible), } \sec\theta = -\frac{3}{2}$$

$$\tan\theta = \pm\sqrt{\frac{9}{4} - 1} = \pm\frac{\sqrt{5}}{2}$$

$$(\because \text{slope is positive} \Rightarrow \tan\theta = -\frac{\sqrt{5}}{2})$$

$$\text{Hence the required equation be } -\frac{3x}{2 \times 3} + \frac{y\sqrt{5}}{2 \times 2} = 1$$

$$\Rightarrow 2x - \sqrt{5}y + 4 = 0$$

Paragraph

The circle $x^2 + y^2 - 8x = 0$ and hyperbola $\frac{x^2}{9} - \frac{y^2}{4} = 1$ intersect at the points A and B.

\n

Q63. Equation of the circle with AB as its diameter is

$$(A) x^2 + y^2 - 12x + 24 = 0 \quad (B) x^2 + y^2 + 12x + 24 = 0 \quad (C) x^2 + y^2 + 24x - 12 = 0$$

$$(D) x^2 + y^2 - 24x - 12 = 0$$

Ans. A

Sol. Solving (a) & (b) for x, we get

$$x = 6$$

$$y = \pm 2\sqrt{3}$$

$$(x - 6)^2 + y^2 - 12 = 0$$

$$x^2 + y^2 - 12x + 24 = 0$$

Option (A) is correct.

Paragraph

Tangents are drawn to the parabola $y^2 = 4x$ from the point $P(6, 5)$ to touch the parabola at Q and R. C_1 is a circle which touches the parabola at Q and C_2 is a circle which touches the parabola at R. Both the circles C_1 and C_2 pass through the focus of the parabola.

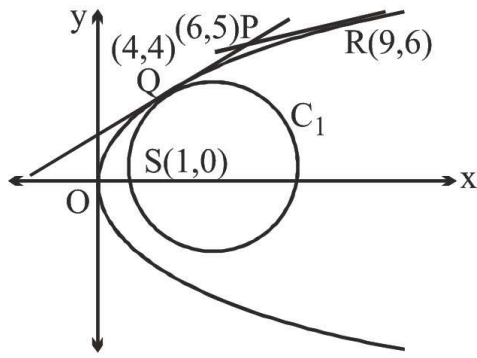
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Q64. Area of the ΔPQR equals

$$(A) 1/2 \quad (B) 1 \quad (C) 2 \quad (D) 1/4$$

Ans. B

Sol.



Equation of tangent of slope m to $y^2 = 4x$ is

$$y = mx + \frac{1}{m}$$

As (1) passes through $P(6, 5)$, so

$$5 = 6m + \frac{1}{m}$$

$$\Rightarrow 6m^2 - 5m + 1 = 0 \Rightarrow m = \frac{1}{2} \text{ or } m = \frac{1}{3}$$

Points of contact are $\left(\frac{1}{m_1^2}, \frac{2}{m_1}\right)$ and $\left(\frac{1}{m_2^2}, \frac{2}{m_2}\right)$

Hence $P(4, 4)$ and $Q(9, 6)$

$$\text{Area of } \triangle PQR = \frac{1}{2}M = \begin{vmatrix} 6 & 5 & 1 \\ 4 & 4 & 1 \\ 9 & 6 & 1 \end{vmatrix} = \frac{1}{2} \Rightarrow (A)$$

Paragraph

Tangents are drawn to the parabola $y^2 = 4x$ from the point $P(6, 5)$ to touch the parabola at Q and R . C_1 is a circle which touches the parabola at Q and C_2 is a circle which touches the parabola at R . Both the circles C_1 and C_2 pass through the focus of the parabola.

\n

Q65. Radius of the circle C_2 is

(A) $5\sqrt{5}$ (B) $5\sqrt{10}$ (C) $10\sqrt{2}$ (D) $\sqrt{210}$

Ans. B

Sol. $y = \frac{1}{2}x + 2 \Rightarrow x - 2y + 4 = 0 \dots\dots(2)$

and $y = \frac{1}{3}x + 3 \Rightarrow x - 3y + 9 = 0$

Now equation of circle C_2 touching $x - 3y + 9 = 0$ at $(9, 6)$, is

$$(x - 9)^2 + (y - 6)^2 + \lambda(x - 3y + 9) = 0$$

As above circle passes through (1, 0), so

$$64 + 36 + 10\lambda = 0 \Rightarrow \lambda = -10$$

$$\text{Circle } C_2 \text{ is } x^2 + y^2 - 28x + 18y + 27 = 0 \quad \dots(3)$$

Radius of C_2 is

$$r_2^2 = 196 + 81 - 27 = 277 - 27 = 250 \Rightarrow r_2 = 5\sqrt{10} \Rightarrow (B)$$

Paragraph

Tangents are drawn to the parabola $y^2 = 4x$ from the point $P(6, 5)$ to touch the parabola at Q and R . C_1 is a circle which touches the parabola at Q and C_2 is a circle which touches the parabola at R . Both the circles C_1 and C_2 pass through the focus of the parabola.

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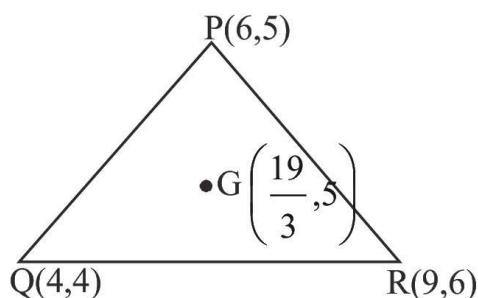
Q66. The common chord of the circles C_1 and C_2 passes through the

(A) incentre of the ΔPQR . (B) circumcentre of the ΔPQR .

(C) centroid of the ΔPQR . (D) orthocentre of the ΔPQR .

Ans. C

Sol.



Equation of C_1

$$(x - 4)^2 + (y - 4)^2 + \lambda(x - 2y + 4) = 0$$

As above circle passes through (1, 0)

$$9 + 16 + \lambda(5) = 0 \Rightarrow \lambda = -5$$

$$\text{Now } C_1 \text{ is } x^2 + y^2 - 13x + 2y + 12 = 0 \quad \dots(4)$$

$\therefore$ Common chord of (3) and (4) is

$$15x - 16y - 15 = 0 \quad \dots(5)$$

Also centroid (G) of ΔPQR is $\left(\frac{19}{3}, 5\right)$

Clearly $\left(\frac{19}{3}, 5\right)$ satisfies equation (5)

Hence (C)]

Paragraph

Let ABCD be a square of side length 2 units. C_2 is the circle through vertices A, B, C, D and C_1 is the circle touching all the sides of the square ABCD. L is a line through A

Q67. If P is a point on C_1 and Q is another point on C_2 , then $\frac{PA^2 + PB^2 + PC^2 + PD^2}{QA^2 + QB^2 + QC^2 + QD^2}$ is equal to

- (A) 0.75 (B) 1.25 (C) 1 (D) 0.5

Ans. A

Sol. Without loss of generality we can assume the square ABCD

with its vertices A(1, 1), B(-1, 1), C(-1, -1), D(1, -1)

P to be the point (0, 1) and Q as $(\sqrt{2}, 0)$

$$\begin{aligned} \text{Then, } & \frac{PA^2 + PB^2 + PC^2 + PD^2}{QA^2 + QB^2 + QC^2 + QD^2} \\ &= \frac{1 + 1 + 5 + 5}{2[(\sqrt{2} - 1)^2 + 1] + 2((\sqrt{2} + 1)^2 + 1)} = \frac{12}{16} = 0.75 \end{aligned}$$

Paragraph

Let ABCD be a square of side length 2 units. C_2 is the circle through vertices A, B, C, D and C_1 is the circle touching all the sides of the square ABCD. L is a line through A

Q68. A circle touches the line L and the circle C_1 externally such that both the circles are on the same side of the line, then the locus of centre of the circle is

- (A) ellipse (B) hyperbola (C) parabola (D) parts of straight line

Ans. C

Sol. Let C' be the said circle touching C_1 and L, so that C_1 and C' are on the same side of L. Let us draw a line T parallel to L at a distance equal to the radius of circle C_1 , on opposite side of L.

Then for N, centre of circle C' , $MN = NO$

$\Rightarrow$ N is equidistant from a line and a point

$\Rightarrow$ locus of N is a parabola.

Paragraph

Let ABCD be a square of side length 2 units. C_2 is the circle through vertices A, B, C, D and C_1 is the circle touching all the sides of the square ABCD. L is a line through A

Q69. A line M through A is drawn parallel to BD. Point S moves such that its distances from the line BD and the vertex A are equal. If locus of S cuts M at T_2 and T_3 and AC

at T_1 , then area of $\Delta T_1 T_2 T_3$ is

- (A) $1/2$ sq. units (B) $2/3$ sq. units (C) 1 sq. unit (D) 2 sq. units

Ans. C

Sol. Since S is equidistant from A and line BD, it traces a parabola. Clearly AC is the axis, A(1, 1) is the focus and $T_1 \left(\frac{1}{2}, \frac{1}{2} \right)$ is the vertex of parabola, $AT_1 = \frac{1}{\sqrt{2}}$

$$T_2 T_3 = \text{latus rectum of parabola} = 4 \times \frac{1}{\sqrt{2}} = 2\sqrt{2}$$

$$\therefore \text{Area} (\Delta T_1 T_2 T_3) = \frac{1}{2} \times 2\sqrt{2} \times \frac{1}{\sqrt{2}} = 1 \text{ sq. units.}$$

Paragraph

Consider the circle $x^2 + y^2 = 9$ and the parabola $y^2 = 8x$. They intersect at P and Q in the first and the fourth quadrants, respectively. Tangents to the circle at P and Q intersect the x-axis at R and tangents to the parabola at P and Q intersect the x-axis at S.

\n

Q70. The ratio of the areas of the triangles PQS and PQR is

- (A) $1 : \sqrt{2}$ (B) $1 : 2$ (C) $1 : 4$ (D) $1 : 8$

Ans. C

$$\text{Sol. } \frac{\text{Ar} \Delta PQS}{\text{Ar} \Delta PQR} = \frac{\frac{1}{2} QP \times ST}{\frac{1}{2} PQ \times TR} = \frac{ST}{TR} = \frac{2}{8} = \frac{1}{4}$$

Paragraph

Consider the circle $x^2 + y^2 = 9$ and the parabola $y^2 = 8x$. They intersect at P and Q in the first and the fourth quadrants, respectively. Tangents to the circle at P and Q intersect the x-axis at R and tangents to the parabola at P and Q intersect the x-axis at S.

\n

Q71. The radius of the circumcircle of the triangle PRS is

- (A) 5 (B) $3\sqrt{3}$ (C) $3\sqrt{2}$ (D) $2\sqrt{3}$

Ans. B

Sol. For ΔPRS ,

$$\text{Ar}(\Delta PRS) = \Delta = \frac{1}{2} SR \times PT = \frac{1}{2} \times 10 \times 2\sqrt{2}$$

$$\therefore \Delta = 10\sqrt{2}, a = PS = 2\sqrt{3},$$

$$b = PR = 6\sqrt{2}, c = SR = 10$$

\therefore \text{ radius of circumference}

$$= R = \frac{abc}{4\Delta} = \frac{2\sqrt{3} \times 6\sqrt{2} \times 10}{4 \times 10\sqrt{2}} 3\sqrt{3}$$

Paragraph

Consider the circle $x^2 + y^2 = 9$ and the parabola $y^2 = 8x$. They intersect at P and Q in the first and the fourth quadrants, respectively. Tangents to the circle at P and Q intersect the x-axis at R and tangents to the parabola at P and Q intersect the x-axis at S.

\n

Q72. The radius of the incircle of the triangle PQR is

- (A) 4 (B) 3 (C) $\frac{8}{3}$ (D) 2

Ans. D

Sol. Radius of incircle

$$= \frac{\text{area of } \triangle PQR}{\text{semi perimeter of } \triangle PQR} = \frac{\Delta}{s}$$

We have $a = PR = 6\sqrt{2}$, $b = QP = PR = 6\sqrt{2}$

$$c = PQ = 4\sqrt{2}$$

$$\text{and } \Delta = \frac{1}{2} PQ \cdot TR = 16\sqrt{2}$$

$$\therefore s = \frac{6\sqrt{2} + 6\sqrt{2} + 4\sqrt{2}}{2} = 8\sqrt{2}$$

$$\therefore r = \frac{16\sqrt{2}}{8\sqrt{2}} = 2$$

Q73. Which of the following equations in parametric form can represent a hyperbolic profile, where 't' is a parameter :

(A) $x = \frac{a}{2} \left(t + \frac{1}{t} \right) \& y = \frac{b}{2} \left(t - \frac{1}{t} \right)$

(B) $\frac{tx}{a} - \frac{y}{b} + t = 0 \& \frac{x}{a} + \frac{ty}{b} - 1 = 0$ (C) $x = e^t + e^{-t} \& y = e^t - e^{-t}$

(D) $x^2 - 6 = 2 \cos t \& y^2 + 2 = 4 \cos^2 \frac{t}{2}$

Ans. A, C, D

Sol. Check option using elimination $\rightarrow$ ACD

Q74. Which of the following equations in parametric form can represent a hyperbolic profile, where 't' is a parameter.

(A) $x = \frac{a}{2} \left(t + \frac{1}{t} \right) \& y = \frac{b}{2} \left(t - \frac{1}{t} \right)$ (B) $\frac{tx}{a} - \frac{y}{b} + t = 0 \& \frac{x}{a} + \frac{ty}{b} - 1 = 0$

(C) $x = e^t + e^{-t}$ & $y = et - e^{-t}$ (D) $x^2 - 6 = 2 \cos t$ & $y^2 + 2 = 4 \cos^2 \frac{t}{2}$

Ans. A, C, D

Sol. -

Q75. Let p and q be non-zero real numbers. Then the equation $(px^2 + qy^2 + r)(4x^2 + 4y^2 - 8x - 4) = 0$ represents

(A) two straight lines and a circle, when $r = 0$ and p, q are of the opposite sign.

(B) two circles, when $p = q$ and r is of sign opposite to that of p .

(C) a hyperbola and a circle, when p and q are of opposite sign and $r \neq 0$.

(D) a circle and an ellipse, when p and q are unequal but of same sign and r is of sign opposite to that of p .

Ans. A, B, C, D

Sol. $(px^2 + qy^2 + r)(4x^2 + 4y^2 - 8x - 4) = 0$
 $\Rightarrow 4x^2 + 4y^2 - 8x - 4 = 0 \Rightarrow (x - 1)^2 + y^2 = 1$

or

$px^2 + qy^2 + r = 0$ will represents

(i) two straight lines if $r = 0$ and p, q are of opposite sign.

(ii) a circle if $p = q$ and r is of opposite sign that of p .

(iii) a hyperbola if p and q are of opposite sign & $r \neq 0$.

(iv) an ellipse if p and q are unequal but of same sign and r is of sign opposite to that of p .

Q76. If the circle $x^2 + y^2 = a^2$ intersects the hyperbola $xy = c^2$ in four points $P(x_1, y_1)$, $Q(x_2, y_2)$, $R(x_3, y_3)$, $S(x_4, y_4)$, then

(A) $x_1 + x_2 + x_3 + x_4 = 0$ (B) $y_1 + y_2 + y_3 + y_4 = 0$ (C) $x_1 x_2 x_3 x_4 = c^4$

(D) $y_1 y_2 y_3 y_4 = c^4$

Ans. A, B, C, D

Sol. solving $xy = c^2$ and $x^2 + y^2 = a^2$

$$x^2 = \frac{c^4}{x^2} = a^2$$

$$x^4 - ax^3 - a^2x^2 + ax + c^4 = 0$$

$$\Rightarrow \sum x_i - ax^3 - a^2x^2 + ax + c^4 = 0$$

$$x^2 + \frac{c^4}{x^2} = a^2$$

$$x^4 - ax^3 - a^2x^2 + ax + c^4 = 0$$

$$\Rightarrow \sum x_i = 0 ; \sum y_i = 0$$

$$x_1 x_2 x_3 x_4 = c^4 \quad \Rightarrow \quad y_1 y_2 y_3 y_4 = c^4$$

Q77. The tangent to the hyperbola, $x^2 - 3y^2 = 3$ at the point $(\sqrt{3}, 0)$ when associated with two asymptotes constitutes

- (A) isosceles triangle which is not equilateral (B) an equilateral triangle
(C) a triangle whose area is $\sqrt{3}$ sq. units (D) a right isosceles triangle.

Ans. B, C

Sol. area of the $\triangle = ab$ sq units ; H : $\frac{\frac{x^2}{3} - y^2}{1} = 1$

Q78. The tangent PT and the normal PN to the parabola $y^2 = 4ax$ at a point P on it meet its axis at points T and N, respectively. The locus of the centroid of the triangle PTN is a parabola whose

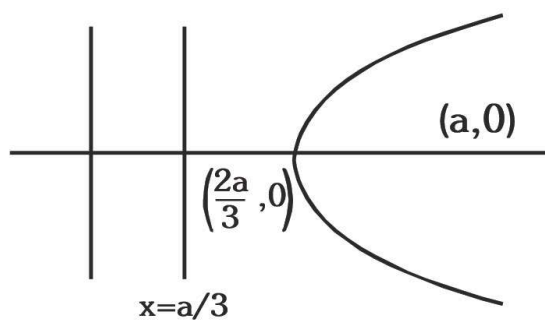
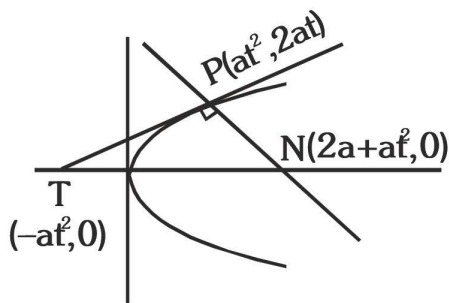
- (A) vertex is $\left(\frac{2a}{3}, 0\right)$ (B) directrix is $x = 0$ (C) latus rectum is $\frac{2a}{3}$
(D) focus is $(a, 0)$

Ans. A, D

Sol. $3h = 2a + at^2$

$$3k = 2at$$

$$3h = 2a + \frac{a \cdot 9k^2}{4a^2}$$



$$y^2 = \frac{4a}{9}(3x - 2a)$$

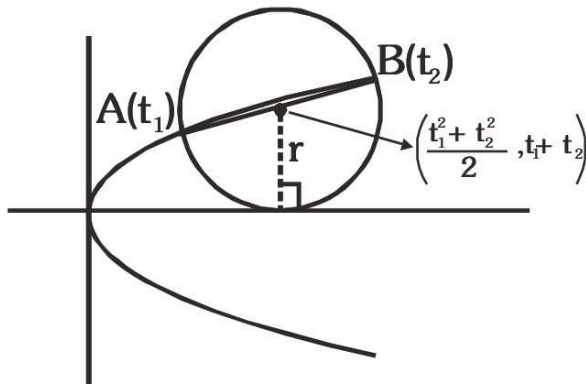
$$y^2 = \frac{4a}{3} \left(x - \frac{2a}{3} \right)$$

Q79. Let A and B be two distinct point on the parabola $y^2 = 4x$. If the axis of the parabola touches a circle of radius r having AB as its diameter, then the slope of the line joining A and B can be

- (A) $-\frac{1}{r}$ (B) $\frac{1}{r}$ (C) $\frac{2}{r}$ (D) $-\frac{2}{r}$

Ans. D, C

Sol.



$$t_1 + t_2 = r$$

$$\frac{2}{r} = \frac{2}{t_1 + t_2}$$

similarly $-\frac{2}{r}$ is also possible

Q80. In a triangle ABC with fixed base BC, the vertex A moves such that $\cos B + \cos C = 4 \sin^2 \frac{A}{2}$. If a, b and c denote the lengths of the sides of the triangle opposite to the angles A, B and C, respectively, then

- (A) $b + c = 4a$ (B) $b + c = 2a$ (C) locus of point A is an ellipse.
(D) locus of point A is a pair of straight lines.

Ans. C, B

Sol. $\cos B + \cos C = 4 \sin^2 \frac{A}{2}$

$$\Rightarrow 2 \cos \left(\frac{B+C}{2} \right) \cdot \cos \left(\frac{B-C}{2} \right) = 4 \sin^2 \frac{A}{2}$$

$$\Rightarrow \cos \left(\frac{B-C}{2} \right) = 2 \sin \left(\frac{A}{2} \right)$$

$$\Rightarrow 2 \frac{A}{2} \cos \frac{B-C}{2} = 4 \sin \frac{A}{2} \cos \frac{A}{2}$$

$$\Rightarrow \sin B + \sin C = 2 \sin A$$

$$\Rightarrow b + c = 2a$$

Q81. An ellipse intersects the hyperbola $2x^2 - 2y^2 = 1$ orthogonally. The eccentricity of the ellipse is reciprocal of that of the hyperbola. If the axes of the ellipse are along the coordinate axes, then

- (A) Equation of ellipse is $x^2 + 2y^2 = 2$ (B) The foci of ellipse are $(\pm 1, 0)$
 (C) Equation of ellipse is $x^2 + 2y^2 = 4$ (D) The foci of ellipse are $(\pm \sqrt{2}, 0)$

Ans. A, B

Sol. $x^2 - y^2 = \frac{1}{2}$ (i) $\rightarrow$ its $e = \sqrt{2}$

e of ellipse is $\frac{1}{\sqrt{2}}$

$\frac{x^2}{2} + \frac{y^2}{1} = b^2$ (ii)

add (i) & (ii) $\frac{3x^2}{2} = \frac{1}{2} = b^2$

$3x^2 = 1 + 2b^2$

$y^2 = \frac{1}{3} + \frac{2b^2}{3} - \frac{1}{6}$

$y^2 = \frac{1}{6}(4b^2 - 1)$

$m_1 \cdot m_2 = -1 \Rightarrow \frac{1 + 2b^2}{3} = \frac{2(4b^2 - 1)}{6}$

$b^2 = 1 \Rightarrow x^2 + 2y^2 = 2.$

Q82. Let the eccentricity of the hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ be reciprocal to that of the ellipse $x^2 + 4y^2 = 4$. If the hyperbola passes through a focus of the ellipse, then

(A) the equation of the hyperbola is $\frac{x^2}{3} - \frac{y^2}{2} = 1$

(B) a focus of the hyperbola is $(2, 0)$ (C) the eccentricity of the hyperbola is $\sqrt{\frac{5}{3}}$

(D) the equation of the hyperbola is $x^2 - 3y^2 = 3$

Ans. D, B

Sol. Given hyperbola is $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$

ellipse is $\frac{x^2}{2^2} + \frac{y^2}{1} = 1$

eccentricity of ellipse $= \sqrt{1 - \frac{1}{4}} = \sqrt{\frac{3}{4}} = \sqrt{\frac{3}{2}}$

eccentricity of hyperbola $= \sqrt{1 + \frac{b^2}{a^2}} = \sqrt{\frac{4}{3}}$

$$\Rightarrow \frac{b^2}{a^2} = \frac{1}{3} \quad \Rightarrow \quad 3b^2 = a^2 \quad \dots\dots(1)$$

also hyperbola passes through foci of ellipse $(\pm\sqrt{3}, 0)$

from (1) & (2)

$$b^2 = 1$$

equation of hyperbola is $\frac{x^2}{3} - \frac{y^2}{1} = 1$

$$\Rightarrow \quad x^2 - 3y^2 = 3$$

$$\text{eccentricity of hyperbola} = \sqrt{1 + \frac{1}{3}} = \sqrt{\frac{4}{3}}$$

$$\text{focus of hyperbola} = \left(\pm \sqrt{3} \frac{2}{\sqrt{3}}, 0 \right) = (\pm 2, 0)$$

Q83. The locus of the mid point of the focal radii of a variable point moving on the parabola, $y^2 = 8x$ is a parabola whose

- (A) Latus rectum is half the latus rectum of the original parabola (B) Vertex is $(1, 0)$
 (C) Directrix is y-axis (D) Focus has the co-ordinates $(2, 0)$

Ans. A, B, C, D

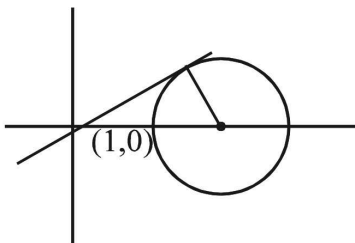
Sol. -

Q84. A variable circle is described to pass through the point $(1, 0)$ and tangent to the curve $y = \tan(\tan^{-1} x)$. The locus of the centre of the circle is a parabola whose

- (A) length of the latus rectum is $2\sqrt{2}$
 (B) axis of symmetry has the equation $x + y = 1$
 (C) vertex has the co-ordinates $(3/4, 1/4)$ (D) directrix is $x - y = 0$

Ans. B, C, D

Sol.

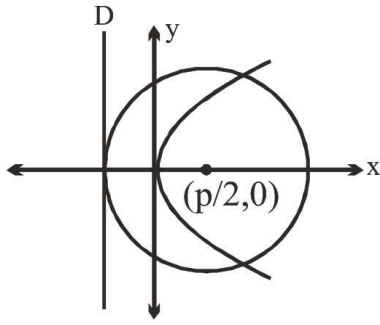


Q85. Consider a circle with its centre lying on the focus of the parabola, $y^2 = 2px$ such that it touches the directrix of the parabola. Then a point of intersection of the circle & the parabola is

- (A) $\left(\frac{p}{2}, p\right)$ (B) $\left(\frac{p}{2}, -p\right)$ (C) $\left(-\frac{p}{2}, p\right)$ (D) $\left(-\frac{p}{2}, -p\right)$

Ans. B

Sol.



Equation of circle will be $\left(x - \frac{P}{2}\right)^2 = y^2 = p^2$

which intersects $y^2 = 2px$

$$\therefore \left(x - \frac{P}{2}\right)^2 + 2px = p^2$$

$$x^2 + px - \frac{3p^2}{4} = 0$$

$$\left(x + \frac{3p^2}{4}\right)\left(x - \frac{p}{2}\right) = 0$$

$$4x^2 + 4px - 3p^2 = 0$$

$$4x^2 + 6px - 2px - 3p^2 = 0$$

$$2x(2x + 3p) - p(2x + 3p)$$

$$(2x + 3p)(2x - p) = 0$$

$$\therefore x = \frac{p}{2} \text{ only}$$

Hence $y^2 = p^2 \Rightarrow y = \pm p \Rightarrow A, B$ **Ans.**

Q86. If from the vertex of a parabola $y^2 = 4x$ a pair of chords be drawn at right angles to one another and with these chords as adjacent sides a rectangle be made, then the locus of the further end of the rectangle is

(A) an equal parabola (B) a parabola with focus at (9, 0)

(C) a parabola with directrix as $x - 7 = 0$

(D) a parabola having tangent at its vertex $x = 8$

Ans. D, C, B, A

Sol. locus is $y^2 = 4a(x - 8a)$

Q87. The focus of the parabola is (1, 1) and the tangent at the vertex has the equation $x + y = 1$. Then :

(A) equation of the parabola is $(x - y)^2 = 2(x + y - 1)$

(B) equation of the parabola is $(x - y)^2 = 4(x + y - 1)$

(C) the co-ordinates of the vertex are $\left(\frac{1}{2}, \frac{1}{2}\right)$ (D) length of the latus rectum is $2\sqrt{2}$

Ans. D, C, B

Sol. -

- Q88. Consider the parabola whose equation is $y = x^2 - 4x$ and the line $y = 2x - b$. Then which of the following is/are correct?
- (A) For $b = 9$ the line is a tangent to the parabola.
- (B) For $b = 7$ the line cuts the parabola in A and B such that the $\angle COD$ is a right angle when 'O' is the origin.
- (C) For some $b \in \mathbb{R}$ the line cuts the parabola in C and D such that x-axis bisects the $\angle COD$.
- (D) For $b > 9$ the line passes outside the parabola.

Ans. A, B, D

Sol. (A) For $b = 9$ solving line and parabola
 $2x - 9 = x^2 - 4x \Rightarrow x^2 - 6x + 9 = 0 \Rightarrow (x - 3)^2 = 0$
 $\therefore$ line is tangent to parabola $\Rightarrow$ (A) is correct

(B) For $b = 7$, line is $y = 2x - 7$
Homogenising parabola with line
 $x^2 - 4x\left(\frac{2x - y}{7}\right) - y\left(\frac{2x - y}{7}\right) = 0$
coefficient of x^2 + coefficient of $y^2 = 0 \Rightarrow 1 - \frac{8}{7} + \frac{1}{7} = 0$
 $\therefore \angle AOB$ is $90^\circ \Rightarrow$ (B) is correct

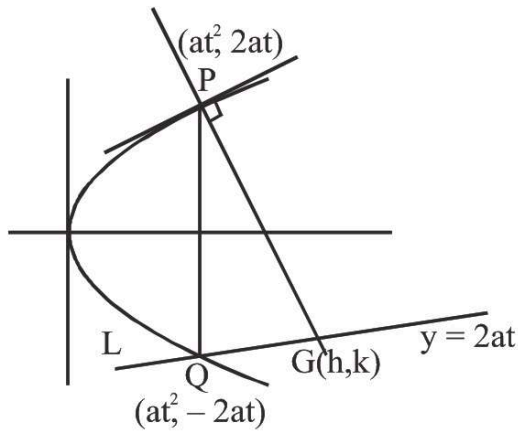
(C) Homogenising the parabola with line, we get
 $x^2 - 4x\left(\frac{2x - y}{b}\right) - y\left(\frac{2x - y}{b}\right) = 0$
for x-axis to bisect $\angle COD$
coefficient of xy should be zero
i.e. $\frac{4}{b} - \frac{2}{b} = 0$
which is not possible for any $b \in \mathbb{R}$. $\Rightarrow$ (C) is false

(D) For $b > 9$
Solving line and parabola
 $x^2 - 6x + b = 0$
 $D = 36 - 4b$
If $b > 9$
then $D < 0$
 $\therefore$ line passes outside the parabola $\Rightarrow$ (D) is correct.]

- Q89. PQ is a double ordinate of the parabola $y^2 = 4ax$. If the normal at P intersect the line passing through Q and parallel to axis of x at G, then locus of G is a parabola with
- (A) length of latus rectum equal to $4a$. (B) vertex at $(4a, 0)$.
- (C) directrix as the line $x - 3a = 0$ (D) focus as $(5a, 0)$

Ans. A, B, C, D

Sol.



$$N : y + tx = 2at + at^3$$

$$L : y = -2at$$

solving together

$$-2at + tx = 2at + at^3$$

$$x = at^2 + 4a$$

$$\text{hence, } h = at^2 + 4a \text{ and } k = -2at$$

$$\frac{k^2}{4a^2} = \frac{h - 4a}{a} \Rightarrow y^2 = 4(x - 4a)$$

Now verify all the alternatives. **Ans.]**

Q90. If θ is eliminated from the equations $a \sec \theta - x \tan \theta = y$ and $\theta \sec \theta + y \tan \theta = x$ (a and b are constant) then the element denotes the equation of

(A) the director circle of the hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$

(B) auxiliary circle of the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$

(C) Director circle of the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$

(D) Director circle of the circle $x^2 + y^2 = \frac{a^2 + b^2}{2}$

Ans. C, D

Sol. $a \sec \theta = y + x \tan \theta$

$b \sec \theta = x - y \tan \theta$

$$(a^2 + b^2) \sec^2 \theta = x^2(1 + \tan^2 \theta) + y^2(1 + \tan^2 \theta)$$

$$\Rightarrow x^2 + y^2 = a^2 + b^2 \Rightarrow \text{(C) and (D)]}$$

Q91. If a hyperbola passes through the focus of the ellipse $\frac{x^2}{25} + \frac{y^2}{16} = 1$ and its transverse and coincides with the major and minor axis of the ellipse, and product of their

eccentricities is 1, then

(A) equation of hyperbola $\frac{x^2}{9} - \frac{y^2}{16} = 1$ (B) equation of hyperbola $\frac{x^2}{9} - \frac{y^2}{25} = 1$

(C) focus of hyperbola (5, 0) (D) focus of hyperbola is $(5\sqrt{3}, 0)$

Ans. A, C

Sol. Eccentricity of ellipse = $\frac{3}{5}$

Eccentricity of hyperbola = $\frac{5}{3}$ and it passes through $(\pm 3, 0)$

$\Rightarrow$ its equation $\frac{x^2}{9} - \frac{y^2}{b^2} = 1$

where $1 + \frac{b^2}{9} = \frac{25}{9} \Rightarrow b^2 = 16$

$\Rightarrow \frac{x^2}{9} - \frac{y^2}{16} = 1 \Rightarrow \frac{x^2}{9} - \frac{y^2}{16} = 1$ and its foci

are $(\pm 5, 0)$

Q92. The equations of common tangents to the parabola $y = x^2$ and $y = -(x - 2)^2$ is/are

(A) $y = 4(x - 1)$ (B) $y = 0$ (C) $y = -4(x - 1)$ (D) $y = -30x - 50$

Ans. A, B

Sol. $x^2 = y$ $(x - 2)^2 = -y$

let slope is n

$y = mn - am^2$ $y = m(x - 2) + \frac{1}{4}m^2$

$y = mn - \frac{m^2}{4}$ $y = mn - 2m + \frac{m^2}{4}$

for common tangent t

$\frac{-m^2}{4} = -2m + \frac{m^2}{4}$

$2m = \frac{2m^2}{4}$

$m = 0,$ $m = 4$

tangents are

$y = 0,$ $y = 4(n - 1)$

Q93. Let $P(x_1, y_1)$ and $Q(x_2, y_2)$, $y_1 < 0$, $y_2 < 0$, be the end points of the latus rectum of the ellipse $x^2 + 4y^2 = 4$. The equations of parabolas with latus rectum PQ are

(A) $x^2 + 2\sqrt{3}y = 3 + \sqrt{3}$ (B) $x^2 - 2\sqrt{3}y = 3 + \sqrt{3}$ (C) $x^2 + 2\sqrt{3}y = 3 - \sqrt{3}$

$$(D) x^2 + 2\sqrt{3}y = 3 - \sqrt{3}$$

Ans. B, C

Sol. Equation of ellipse is $\frac{x^2}{4} + y^2 = 1$

$$\text{eccentricity } e = \frac{\sqrt{3}}{2}$$

so focus are $(\sqrt{3}, 0)$ & $(-\sqrt{3}, 0)$

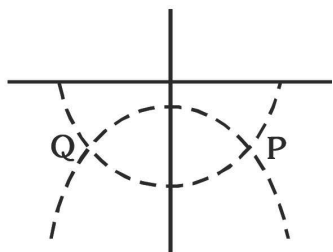
so end points of latus rectum will be

$$\left(\sqrt{3}, \frac{1}{2}\right), \left(\sqrt{3}, -\frac{1}{2}\right), \left(-\sqrt{3}, \frac{1}{2}\right) \& \left(-\sqrt{3}, -\frac{1}{2}\right)$$

$$\because y_1 < 0 \& y_2 < 0$$

Hence coordinates of P & Q will be

$$P\left(\sqrt{3}, \frac{1}{2}\right) \& Q\left(-\sqrt{3}, -\frac{1}{2}\right)$$



So now equation of parabola taking these points as end points of latus rectum.

Focus will be $(0, -\frac{1}{2})$

$$4a = 2\sqrt{3} \Rightarrow a = \frac{\sqrt{3}}{2}$$

Hence vertex of the parabolas will be

$$\left(0, -\frac{1}{2} - \frac{\sqrt{3}}{2}\right)$$

so eq. of parabolas will be

$$x^2 = -3\sqrt{3}\left(y + \frac{1}{2} - \frac{\sqrt{3}}{2}\right) \&$$

$$x^2 = 2\sqrt{3}\left(y + \frac{1}{2} + \frac{\sqrt{3}}{2}\right)$$

$$x^2 + 2\sqrt{3}y = 3 - \sqrt{3}$$

$$x^2 - 2\sqrt{3}y = 3 + \sqrt{3}$$

Q94. **Column-I**

Column-II

(A) The number of real common tangents to the circle $5x^2 + 5y^2 = 16$

- (P) 1
and the hyperbola $3x^2 - y^2 = 48$, is
- (B) The number of real common normals to parabola $y^2 = 4x$
- (Q) 2
and the circle $(x - 1)^2 + (y - 1)^2 = 1$, is
- (C) If P is any point on ellipse $\frac{x^2}{4} + 4y^2 = 1$ whose foci are at A and B, (R)
3
then the maximum value of (PA)(PB) equals
- (S) 4
- (D) The length of latus-rectum of the parabola defined parametrically
(T) 5
by equations $x = \cos t - \sin t$ and $y = \sin 2t$ equals
- (E) The value of a for the ellipse $= 1$ ($a > b$), if the extremities of the latus-rectum of the ellipse having positive ordinate lies on the parabola $x^2 = -2(y - 2)$, is

Ans. (A) S ; (B) P ; (C) S ; (D) P ; (E) Q

Sol. (A) $S : x^2 + y^2 = \frac{16}{5}$

$$H : \frac{x^2}{16} - \frac{y^2}{48} = 1 \quad (\text{dividing by } 48)$$

tangent to H is, $y = mx \pm \sqrt{16m^2 - 48}$

use $p = r$

$$\frac{\sqrt{16m^2 - 48}}{\sqrt{1 + m^2}} = \frac{4}{\sqrt{5}} \Rightarrow 5(16m^2 - 48) = 16(1 + m^2) \Rightarrow 64m^2 = 16 \times 16$$

$$m^2 = 4 \Rightarrow m = \pm 2$$

$\Rightarrow$ 4 common tangent $\Rightarrow$ (S) [for each value of m there are two parallel tangents]

(B) $P : y^2 = 4x$

$$S : (x - 1)^2 + (y - 1)^2 = 1$$

normal to P: $y = mx - 2m - m^3$ passes through the centre of the circle (1, 1)

$$1 = m - 2m - m^3$$

$$m^3 + m + 1 = 0 \Rightarrow f'(m) > 0 \Rightarrow \text{one real root}$$

$\therefore$ one common normal $\Rightarrow$ (P)

(C) We have $PA + PB = 4$

Now $\frac{PA + PB}{2} \geq \sqrt{PA \cdot PB} \Rightarrow PA \cdot PB \leq 4$ **Ans.**

(D) We have $x = \cos t - \sin t$, $y = \sin 2t$

$$\Rightarrow x^2 = 1 - y \Rightarrow x^2 = -(y - 1)$$

Clearly $1(\text{L.R}) = 1$ **Ans.**

(E) $\left(\pm ae, \frac{b^2}{a}\right)$ are extremities of the latus-rectum having positive ordinates.

$$\Rightarrow a^2 e^2 = -2\left(\frac{b^2}{a} - 2\right) \quad \dots(1)$$

$$\text{But } b^2 = a^2 (1 - e^2) \quad \dots(2)$$

$$\therefore \text{From (1) and (2), we get } a^2 e^2 - 2ae^2 + 2a - 4 = 0 \Rightarrow ae^2 (a - 2) + 2(a - 2) = 0$$

$$\therefore (ae^2 + 2)(a - 2) = 0$$

Hence $a = 2$.

Q95. Match the column Type Question

Identify the conic whose equations are given in column-I.

Column-I	Column-II
(Equation of a conic)	(Nature of conic)
(A) $xy + a^2 = a(x + y)$	(P) Ellipse
(B) $2x^2 - 72xy + 23y^2 - 4x - 28y - 48 = 0$	(Q) Hyperbola
(C) $6x^2 - 5xy - 6y^2 + 14x + 5y + 4 = 0$	(R) Parabola.
(D) $14x^2 - 4xy + 11y^2 - 44x - 58y + 71 = 0$	(S) line pair
(E) $4x^2 - 4xy + y^2 - 12x + 6y + 9 = 0$	
(A) (A) S, (B) Q, (C) S, (D) P, (E) R	(B) (A) R, (B) Q, (C) S, (D) P, (E) S
(C) (A) S, (B) Q, (C) P, (D) S, (E) R	(D) (A) S, (B) S, (C) Q, (D) R, (E) P

Ans. 5f06b350b58d37a1927fe540

Sol. -

Q96. Column-I

Column-II

(A) The eccentricity of the ellipse which meets the straight line

(P) $\frac{1}{2}$

$2x - 3y = 6$ on the X-axis and the straight line $4x + 5y = 20$

on the Y-axis and whose principal axes lie along the coordinate axes, is

(B) A bar of length 20 units moves with its ends on two fixed straight lines

(Q) $\frac{1}{\sqrt{2}}$

at right angles. A point P marked on the bar at a distance of 8 units from one end describes a conic whose eccentricity is

(C) If one extremity of the minor axis of the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ and

(R) $\frac{\sqrt{5}}{3}$

the foci form an equilateral triangle, then its eccentricity, is

(D) There are exactly two points on the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$

(S) $\frac{\sqrt{7}}{4}$

whose distance from the centre of the ellipse are greatest and

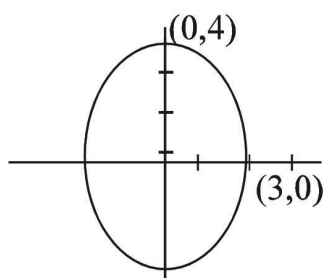
equal to $\sqrt{\frac{a^2 + 2b^2}{2}}$. Eccentricity of this ellipse is equal to

(A) (A) R; (B) S ; (C) P ; (D) Q (B) (A) S; (B) R ; (C) P ; (D) Q

(C) (A) S; (B) R ; (C) Q ; (D) P (D) (A) Q; (B) P ; (C) R ; (D) S

Ans. 5f06b81990e4b0a1523b17a0

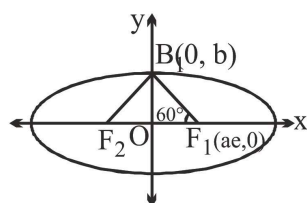
Sol.



(A)

$$e^2 = 1 - \frac{9}{16} = \frac{7}{16} \text{ ellipse touch the x-axis at } (3, 0) \text{ and y-axis at } (0,4)$$

(B) Locus is as ellipse



(C)

$$\text{We have } \frac{b}{ae} = \sqrt{3} \Rightarrow a^2(1 - e^2) = 3a^2e^2 \text{ (As } b^2 = a^2(1 - e^2))$$

$$\Rightarrow e^2 = \frac{1}{4}$$

$$\text{Hence } e = \frac{1}{2} \text{ Ans.}$$

(D) The given distance is clearly the length of semi major axis

$$\text{Thus, } \sqrt{\frac{a^2 + 2b^2}{2}} = a \Rightarrow 2b^2 = a^2 \Rightarrow 2a^2(1 - e^2) = a^2$$

$$\Rightarrow e^2 = \frac{1}{2} \quad e = \frac{1}{\sqrt{2}} \text{ Ans.}]$$

Q97. Column-I

Column-II

(A) If the chord of contact of tangents from a point P to the

(P)

Straight line

parabola $y^2 = 4ax$ touches the parabola $x^2 = 4by$, the locus of P is

(B) A variable circle C has the equation (Q)

Circle

$x^2 + y^2 - 2(t^2 - 3t + 1)x - 2(t^2 + 2t)y + t = 0$, where t is a parameter.

The locus of the centre of the circle is

(C) The locus of point of intersection of tangents to an ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ (R)

Parabola

at two points the sum of whose eccentric angles is constant is

(D) An ellipse slides between two perpendicular straight lines. (S)

Hyperbola

Then the locus of its centre is

(A) (A) R; (B) S; (C) P; (D) Q (B) (A) S; (B) R; (C) Q; (D) P

(C) (A) S; (B) R; (C) P; (D) Q (D) (A) Q; (B) P; (C) R; (D) S

Ans. 5f06bbe9b58d37a1927fe5a1

Sol. (A) $yy_1 = 2a(x + x_1)$; $x^2 = 4by = 4b\left[\left(\frac{2a}{y_1}\right)(x + x_1)\right] \Rightarrow y_1x^2 - 8abx - 8abx_1 = 0$;

$D = 0$ gives $xy = -2ab \Rightarrow$ Hyperbola

(B) centre is $h = t^2 - 3t + 1$ (1)

$k = t^2 + 2t$ (2)

(2) - (1) gives $-x + y = 5t - 1$

or $t = \frac{1 - x + y}{5}$

Substituting the value of t in (2)

$$y = \left(\frac{y - x + 1}{5}\right)^2 + 2\left(\frac{y - x + 1}{5}\right)$$

$$25y = (y - x + 1)^2 + 10(y - x + 1)$$

$$25y = y^2 + x^2 + 1 - 2xy - 2x + 2y + 10y - 10x + 10$$

$$x^2 + y^2 - 2xy - 12x - 13y + 11 = 0$$

which is a parabola

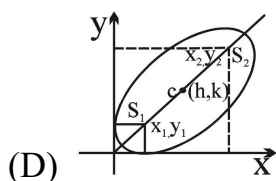
as $\Delta \neq 0$ and $h^2 = ab$

$$(C) h = \frac{a \cos \frac{\alpha + \beta}{2}}{\cos \frac{\alpha - \beta}{2}}; k = \frac{b \sin \frac{\alpha + \beta}{2}}{\cos \frac{\alpha - \beta}{2}}$$

given $\frac{\alpha + \beta}{2} = \text{constant} = C$

$$\therefore \cos \frac{\alpha - \beta}{2} = \frac{a \cos C}{h} = \frac{b \sin C}{k} \Rightarrow y = \left(\frac{b}{a} \tan C\right)x$$

Locus of (h, k) is a straight line



$$y_1 y_2 = x_1 x_2 = b^2 \quad \dots(1)$$

$$\text{and } (x_2 - x_1)^2 + (y_2 - y_1)^2 = 4(a^2 - b^2) \quad \dots(2)$$

$$\text{Also } 2h = x_1 + x_2$$

$$2k = y_1 + y_2$$

$$\text{from (2) } (x_1 + x_2)^2 + (y_1 + y_2)^2 - 4(x_1 x_2 + y_1 y_2)$$

$$= 4(a^2 - b^2) \quad (h^2 + k^2) - (2b^2) = (a^2 - b^2)$$

$$\therefore x^2 + y^2 = a^2 + b^2 \Rightarrow \text{Circle}$$

Alternative: Equation of director circle with centre (h, k)

$$(x - h)^2 + (y - k)^2 = a^2 + b^2$$

$$(0, 0) \text{ lies on it } \Rightarrow h^2 + k^2 = a^2 + b^2 \Rightarrow \text{locus is } x^2 + y^2 = a^2 + b^2.$$

Q98.

Column-I

Column-II

(A) For an ellipse $\frac{x^2}{9} + \frac{y^2}{4} = 1$ with vertices A and A', tangent drawn at the

(P) 2

point P in the first quadrant meets the y-axis in Q and the chord A'P meets the y-axis in M. If 'O' is the origin then $OQ^2 - MQ^2$ equals to

(B) If the product of the perpendicular distances from any point on the

(Q) 3

hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ of eccentricity $e = \sqrt{3}$ from its asymptotes

is equal to 6, then the length of the transverse axis of the hyperbola is

(C) The locus of the point of intersection of the lines

(R) 4

$$\sqrt{3}x - y - 4\sqrt{3}t = 0 \text{ and } \sqrt{3}tx + ty - 4\sqrt{3} = 0$$

(where t is a parameter) is a hyperbola whose eccentricity is

(D) If F_1 & F_2 are the feet of the perpendiculars from the foci S_1 & S_2

(S) 6

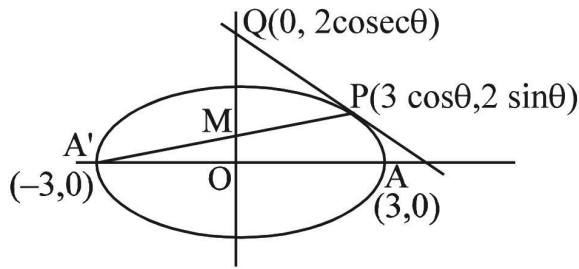
of an ellipse $\frac{x^2}{6} + \frac{y^2}{4} = 1$ on the tangent at any point P on the ellipse, then $(S_1 F_1) \cdot (S_2 F_2)$ is equal to

(A) (A) R; (B) S; (C) P; (D) Q (B) (A) S; (B) R; (C) P; (D) Q

(C) (A) R; (B) S; (C) Q; (D) P (D) (A) Q; (B) P; (C) S; (D) R

Ans. 5f06c162b58d37a1927fe636

Sol.



(A)

$$a = 3 ; b = 2$$

$$T : \frac{x \cos \theta}{3} + \frac{y \sin \theta}{2} = 1$$

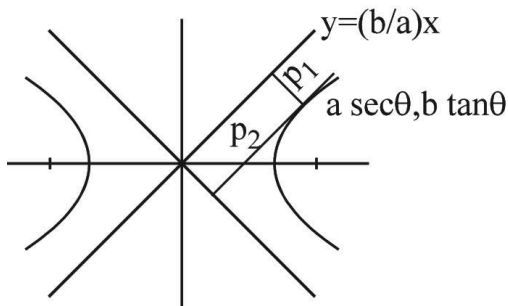
$$x = 0; y = 2 \operatorname{cosec} \theta$$

$$\text{chord A'P, } y = \frac{2 \sin \theta}{3(\cos \theta + 1)}(x + 3)$$

$$\text{put } x = 0 \quad y = \frac{2 \sin \theta}{1 + \cos \theta} = OM$$

$$\text{Now } OQ^2 - MQ^2 = OQ^2 - (OQ - OM)^2 = 2(OQ)(OM) - OM^2 = OM \{ 2(OQ) - (OM) \}$$

$$= \frac{2 \sin \theta}{1 + \cos \theta} \left[\frac{4}{\sin \theta} - \frac{2 \sin \theta}{1 + \cos \theta} \right] = \frac{4 \sin \theta}{1 + \cos \theta} \left[\frac{2(1 + \cos \theta) - (1 - \cos^2 \theta)}{\sin \theta(1 + \cos \theta)} \right] = \frac{4(1 + \cos \theta)(2 - 1 + \cos \theta)}{(1 + \cos \theta)(1 + \cos \theta)}$$



(B)

$$p_1 p_2 = \frac{a^2 b^2}{a^2 b^2} = \frac{a^2 \cdot a^2 (e^2 - 1)}{a^2 e^2} = 6$$

$$\frac{2a^2}{3} = 6 \Rightarrow a^2 = 9 \Rightarrow a = 3$$

$$\text{hence } 2a = 6$$

$$(C) \text{ hyperbola } \frac{x^2}{16} - \frac{y^2}{48} = 1$$

(D) Product of the feet of the perpendiculars is equal to the square of its semi minor axis

Q99. Match the following

Normals are drawn at points P, Q and R lying on the parabola $y^2 = 4x$ which intersect at (3, 0). Then

Column-I

Column-II

(A) Area of ΔPQR

(P) 2

(B) Radius of circumcircle of ΔPQR

(Q) 5/2

(C) Centroid of $\triangle PQR$

(R) $(\frac{5}{2}, 0)$

(D) Circumcentre of $\triangle PQR$

(S) $(\frac{2}{3}, 0)$

(A) (A) Q, (B) P, (C) S, (D) R (B) (A) P, (B) Q, (C) S, (D) R

(C) (A) P, (B) Q, (C) R, (D) S (D) (A) R, (B) S, (C) Q, (D) P

Ans. 5f06c2a7992a70a16d4b3c24

Sol. Equation of normal

$$y = mx - 2am - am^3$$

$$a = 1$$

passing $(3, 0)$

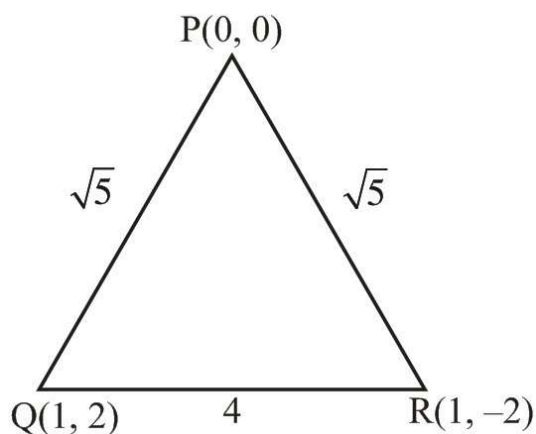
$$0 = 3m - 2m - m^3$$

$$m - m^3 = 0$$

$$m = 0, \quad m^2 = 1$$

$$m = \pm 1$$

$P(0, 0)$ $Q(1, 2), R(1, -2)$



$$\Delta = \begin{vmatrix} 0 & 0 & 1 \\ 1 & 2 & 1 \\ 1 & -2 & 1 \end{vmatrix}$$

$$= \frac{1}{2} |(-2 - 2)| = 2$$

$$R = \frac{abc}{4\Delta} = \frac{20}{2(4)} = \frac{5}{2}$$

centroid $(\frac{2}{3}, 0)$

Circum centre $(\frac{5}{2}, 0)$

Q100. Match the statements in Column I with the properties in Column II.

Column I

- (A) Two intersecting circles
- (B) Two mutually external circles
- (C) Two circles, one strictly inside the other
- (D) Two branches of a hyperbola

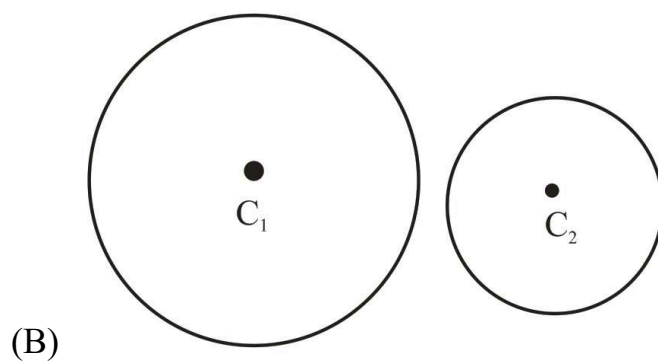
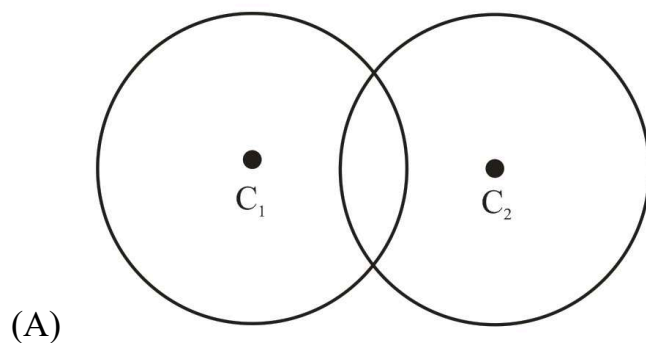
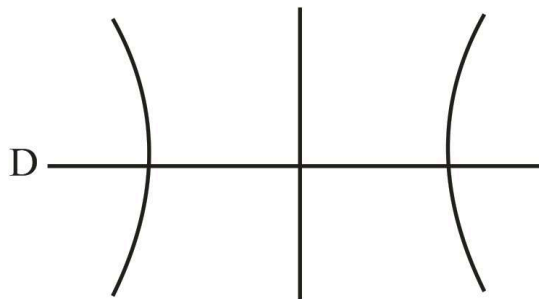
Column II

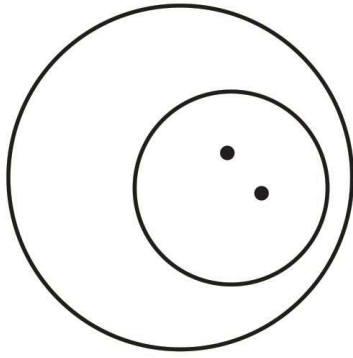
- (P) have a common tangent
- (Q) have a common normal
- (R) do not have a common tangent
- (S) do not have a common normal

- (A) (A) P, Q; (B) P, Q; (C) Q, R; (D) Q, R
- (B) (A) P, R; (B) P, R; (C) Q, R; (D) Q, R
- (C) (A) P, Q; (B) P, Q; (C) Q, S; (D) Q, S
- (D) (A) P, S; (B) P, S; (C) Q, R; (D) Q, R

Ans. 5f06c3e9b58d37a192803b6e

Sol.





(C)